



<Customer Name> Oracle Database@Azure

Solution Definition

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Guide:

Author Responsibility

- Chapter 1-3: Sales Consultant
- Chapter 4: Implementer

Document Control

Guide:

The first chapter of the document describes the metadata for the document. Such as versioning and team members.

1.1 Version Control

Guide:

A section describing the versions of this document and its changes.

Example:

Version	Authors	Date	Comments
1.0	Base Template	1st June 2023	Created a new Solution Definition document. To be used for iterative review and improvement.
1.1	Base Template	1st July 2023	Update Template per feedback. Added security-templated texts and annex.
1.2	Base Template	1st August 2023	Update Template per feedback. As per Confluence.
2.0	Base Template	1st September 2023	Added Networking Annex
2.1	Base Template	1st September 2023	Updated LZ Snippet Added 'Base Template' to the version table instead of 'Name Surname'
2.2	Base Template	16th October 2023	Upgraded the Logical Architecture as mandatory. It is now included in the 'Mandatory' template.
2.3	Base Template	16th January 2024	Added comment for solution snippets Updates Acronyms
2.4	Base Template	26th February 2024	Added the network firewall in the requirement, the solution considerations, and in the Annex.
2.5	Base Template	8th April 2024	Added 'manageability' in the requirement, the solution considerations, and in the Annex.
2.6	Base Template	26th February 2025	Removed 'ACME', added BoM example, updated NFR's, consolidated Solution Considerations, consolidated Business Context, added Onboarding, and replaced 'Workload' with 'Solution'.
2.7	OD@Azure Template	22nd August 2025	Added specific OD@Azure considerations to standard 'Workload to OCI' template version 2.6 (incorporating all elements from previous latest OD@Azure template 2.6.3)

1.2 Team

Guide:

A section describing the team.

Example:

Name	Email	Role	Company
Name Surname	example@example.com	Tech Solution Specialist	Oracle
Ada Lovelace	example@example.com	Account Cloud Engineer	Oracle

1.3 Document Purpose

Guide:

Describe the purpose of this document and the Oracle-specific terminology

Example:

This document provides a high-level solution definition for the Oracle solution and aims at describing the current state, and to-be state as well as a potential high-level project scope and timeline for <Service Provider>.

The document may refer to a 'Workload', which summarizes the full technical solution for a customer (You) during a single engagement. The Workload is described in the chapter [Workload Requirements and Architecture](#).

This is a living document, additional sections will be added as the engagement progresses resulting in a final Document to be handed over to the <Service Provider>.

Business Context

Guide:

Describe the customer's business and background. What is the context of the customer's industry and LoB? What are the business needs and goals that this solution is an enabler for? How does this technical solution impact and support the customer's business goals? Does this solution support a specific customer strategy, or maybe certain customer values? How does this solution help our customers to either generate more revenue or save costs?

Describe a clear statement of a business value as part of the full solution scope. Try to keep it SMART: Specific, Measurable, Assignable, Realistic, and Time-Related - Agree on the business value with the customer. Keep it business-focused, and speak the language of the LoB: "Increase Customer Retention by 3% in the next year" or "Grow Customer Base with Executive Decision-Making from our Sales and Support Data". Avoid technical success criteria such as "Migrate App X to Oracle Cloud" or "Provision 5 Compute Instances". Avoid Oracle success criteria and language "Get Solution Consuming on OCI".

Furthermore, highlight Oracle differentiator and key values of the solution of our solution for the customer, allowing the customer to make decisions quickly.

Example:

[customer] is a large airline operator, looking to modernize their application platform. Through this modernization, [customer] aims to improve the customer experience, enhance performance by 30%, cope with the larger forecasted transaction volume, and simplify future expansion into additional geographies.

[customer] has in the last year initiated a program to move all existing IT assets to public cloud where possible. As part of this program, a decision has been made to use Azure as the main/primary public cloud provider.

When the decision was made to move these workloads to Azure, it was vital to maintain the best performance for the Oracle databases. Therefore [customer] plans to use Oracle Database@Azure. This service provides the power of Exadata without investing in hardware

and without having to manage the infrastructure but only the databases. It provides this as an OCI service, situated in an Azure data center and natively integrated in the Azure network.

Solution Requirements and Architecture

3.1 Overview

Guide:

Describe the Solution: What applications and environments are part of this migration or new implementation project, and what are their names? A possible implementation will be scoped later and can be a subset of the Solution Definition and proposed overall solution. For example, a solution could exist of two applications, but the implementer would only include one environment of one application. The solution chapter is about the whole solution and the implementation scope will be described later in the chapter [Solution Scope](#).

You can describe an existing solution in the optional chapter for a Current State Architecture , which is typically located after the Non functional requirements and before the Future State Architecture.

Example:

Currently, the database environment is deployed on a single Exadata system, while the application servers for both the ticketing application and BI application are deployed on a virtual machines within a VMWare estate. The batch processes that update the DWH database run as custom Python programs on number of Linux hosts, also hosted on the same VMWare estate.

Below is the list of databases with their existing known versions, platforms, size, life cycle environment and virtual cluster.

Database name	Version	DB Platform	DB Type	DB Size	Env.	Cluster	Requirements
OLPT	19.19	Exadata (Linux x86-64)	Single PDB in CDB	1.2TB	Prod	clu-oltp-prod	Upgrade to 19.23. Keep separate cluster and CDB as currently. DR to be set up on separate Exadata Infrastructure.
DWH	19.19	Exadata (Linux x86-64)	Single PDB in CDB	7TB	Prod	clu-dwh-prod	Upgrade to 19.23. Keep separate cluster and CDB as currently. DR to be set up on separate Exadata Infrastructure.
OLPT-TEST	19.19	Exadata (Linux x86-64)	Single PDB in CDB	700GB	Non-Prod	clu-oltp-nonprod	Upgrade to 19.23, move PDB to same CDB as OLTP-DEV. Keep separate cluster from Prod and DWH resources
OLPT-DEV	19.19	Exadata (Linux x86-64)	Single PDB in CDB	500GB	Non-Prod	clu-oltp-nonprod	Upgrade to 19.23, move PDB to same CDB as OLTP-TEST. Keep separate cluster from Prod and DWH resources
DWH-TEST	19.19	Exadata (Linux x86-64)	Single PDB in CDB	3TB	Non-Prod	clu-dwh-nonprod	Upgrade to 19.23, move PDB to same CDB as DWH-DEV. Keep separate cluster from Prod and OLTP resources
DWH-DEV	19.19	Exadata (Linux x86-64)	Single PDB in CDB	1TB	Non-Prod	clu-dwh-nonprod	Upgrade to 19.23, move PDB to same CDB as DWH-TEST. Keep separate cluster from Prod and OLTP resources

3.2 Non-Functional Requirements

Guide:

Describe the high-level technical requirements for the solution. Consider all sub-chapters, but decide and choose which Non-Functional Requirements are necessary for your engagement. You might not need to capture all requirements for all sub-chapters.

This chapter is for describing customer-specific requirements (needs), not to explain Oracle solutions or capabilities.

3.2.1 Regulations and Compliances Requirements

Guide:

This section captures specific regulatory or compliance requirements for the solution. These may limit the types of technologies that can be used and may drive some architectural decisions.

The [Oracle Cloud Infrastructure Compliance Documents service](#) lets you view and download compliance documents.

However, the Oracle Database@Azure solution provides a part of OCI inside Azure. Therefore any compliance documents relevant to either Azure or OCI do not automatically apply to OD@Azure. It is vital that any requirements are captured so that they can be checked against assessments already done, or on the roadmap. Current Oracle Database@Azure compliance information can be found in [the documentation](#)

What needs to be captured are any requirements of the OD@Azure elements, and anything running in OCI. Requirements for specific elements of the solution that will be running completely in Azure (for example AVS, or Azure App Service) do not have to be captured here.

If there are none, then please state it. Leave the second sentence as a default in the document.

Example:

At the time of this document creation, no Regulatory and Compliance requirements have been specified.

In addition to these requirements, the latest version of the [CIS Oracle Cloud Infrastructure Foundation Benchmark](#) will be applied to the Customer tenancy.

3.2.2 Environments required in Azure

Guide:

A diagram or list detailing all the required environments (e.g. development, test, live, production, etc.). Focus on the database@Azure/OCI elements, but also keep in consideration how they interact with the rest of the environments in Azure.

As a minimum we should capture the Exadata hardware requirements of OD@Azure in this section.

If you like to describe a current state, you can use or add the chapter 'Current Sate Architecture' before the 'Future State Architecture'.

Example:

Production and DR will each require a separate 1/4 rack Exadata in different availability zones within the same Azure region Germany West Central (linked to OCI region Frankfurt). Dev and Test are co-located on another 1/4 rack Exadata.

The Production and DR exadata infrastructures will host different VM clusters for each of the databases. The Dev/Test databases will share clusters between workload types.

Name	Size of Prod	Location	DR	Scope
Production	100% (1 x 1/4 rack Exadata X11M, 880 ECPU)	Azure Germany West Central	Yes	In Scope
DR	50%	Azure Germany West Central	No	In Scope
Dev & Test	25%	Azure Germany West Central	No	In Scope

3.2.3 Reliability and Resilience

Guide:

This section captures the resilience and recovery requirements for the solution. Note that these may be different from the current system.

The Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirement of each environment should be captured in the environments section above, and wherever possible.

- *What are the RTO and RPO requirements of the Application?*
- *What are the SLAs of the application?*
- *What are the backup requirements*

Note that if needed, this section may also include an overview of the proposed backup and disaster recovery proposed architectures.

Note that special consideration needs to be given for to the specifics of HA/DR in OD@Azure. Specifically:

- *The regional availability of the solution. See 'Where can I use the service?' in the [FAQ](#)*
- *Differences in naming conventions (Availability Zones in Azure rather than the OCI Availability Domains)*
- *The availability of the solution within only 2 AZs in each region (for multi-AZ regions)*

Due to the Multicloud nature of the OD@Azure solution, there can be considerable traffic between Azure and OCI datacenters in Disaster Recovery scenarios. For example:

- *Under normal circumstances, Dataguard network traffic between databases is advised to run through OCI.*
 - *Whilst it is possible to run this traffic through Azure only, all reference architectures are based on the traffic going through OCI. For performance and reliability reasons it should be highly encouraged to stick to this model. See architectures in [the documentation](#). Specifically refer to Figure 34-2 (MAA Gold DR Architecture in the Same Azure Region) and Figure 34-3 (MAA Gold DR Architecture Across Two Azure Regions), showing connectivity of the networks through OCI rather than Azure.*
- *It is possible for backups to be stored in OCI Object Storage or Autonomous Recovery Service (RCV).*
 - *The recommended choice should always be to leverage Autonomous Recovery Service for backups. This is available either @OCI or @Azure, the customer has the choice to leverage RCV in either Azure or OCI.*
 - *Object Storage is always @OCI.*

If there are any (security) requirement to keep all data within the Azure data centers at all times, then DR and Backup solutions need to be considered with some care, and this would have to be captured in this section.

This chapter is mandatory, while there could be no requirements on HA/DR, please mention that in a short single sentence.

Example:

The databases will be implemented on 2-node Exadata clusters, thereby providing high availability. Oracle MAA Best Practices and Reference Architectures should be considered for all environments.

Both applications have a RPO of 0, with an RTO of 5 minutes. For this reason Active Data Guard will be used to keep the Production and DR databases in sync. The traffic between the Primary and Standby databases will run through the Oracle-Managed network.

In addition, Oracle Autonomous Recovery service with Real-time data protection (ie Zero Data Loss mode) in the Azure data center will be used to accommodate increasing demands for Ransomware Protection (Compliance and Immutability) and Zero RPO.

3.2.4 Security Requirements

Guide:

Capture the Non-Functional Requirements for security-related topics. The requirements can be (but don't have to be) separated into:

- *Identity and Access Management*

- *Data Security*

Other security topics, such as network security, application security, key management, or others can be added if needed.

Take note that parts of the solution running on Azure but which are not directly related to Oracle Database@Azure, are not required to be captured here. For example Azure application security does not have to be captured here.

For Oracle Database@Azure there are several specific considerations which can be clarified during this phase, or later:

- *OCI access for (Database) Management, preferably to be federated to Azure Entra ID, or any alternative the customer is using.*
- *OD@Azure onboarding requirements (required privileges)*
- *Azure governance: Capture Azure subscription design, resource groups and users/groups details. E.g. current application environments subscriptions/resource group design to be used with OD@Azure deployment. Take care to compare with limitations of OD@Azure in terms of mapping to subscriptions, resource groups, etc.*
- *IAM*

3.2.4.1 Identity and Access Management

Example:

[customer] has an existing IAM setup based on Azure Entra ID. OCI IAM will be federated with Azure Entra ID.

The OD@Azure service will automatically create groups with administrative privileges, which will be assigned to the correct admins.

Additionally, all resources for these workloads will run in the same Azure subscription. There will be different resource groups for the production, DR and non-production workloads.

3.2.4.2 Data Encryption / Key Management

For Key Management, if customer-managed keys are required, then additional consideration needs to be given to the implication of choosing between the options:

- *Oracle Key Vault, running on Azure VMs. This option is normally positioned as the default 'right' choice unless there are specific reasons not to use this.*
- *Cloud-based services: OCI Vault (running in OCI) or Azure Key Vault (AKV). These provide less flexibility than OKV, but in specific circumstances they can be the right choice, for example in solutions where the keys need to be held in hardware.*
- *File based. Unless this is a very simple solution (or a trial/POC), file based key management is not recommended.*

Example:

[customer] has a requirement to manage their own Oracle Database@Azure encryption keys. For this there are 3 options:

- OCI Vault
- Azure Key Vault
- Oracle Key Vault (OKV)

Oracle strongly recommends the use of OKV for the key management of Oracle Database@Azure, which is indeed what will be used for this deployment.

3.2.5 Networking Requirements

Guide

Capture the Non-Functional Requirements for networking-related topics as it relates to how Oracle Database@Azure communicates with the rest of the Azure network.

- *Understand Azure Network topology with existing and new requirements.*

- *How should the application connect to the database (same VNet, Hub & Spoke peered VNets, Virtual WAN)?*
- *Connectivity from other locations (on premises, other Azure regions, other clouds, etc)*
- *Any requirement of using private endpoints*
- *DNS requirements (managed, or manually integrated)*
- *Latency requirements (both from Azure to OD@Azure, and if required, from OCI to OD@Azure).*

If the OCI network elements of Oracle Database@Azure are to be leveraged in this workload, then standard OCI network requirements can also be described here, and you can use the networking questions in the [Annex](#)

As businesses increasingly rely on Cloud Infrastructure to store, process, and transmit sensitive data, the need for comprehensive security solutions has never been more important. Potential customers evaluating network security solutions typically prioritize the following requirements: Some of the broader category considerations are below.

- *Data Protection: Safeguarding sensitive information against unauthorized access, theft, or modification is a primary concern for any organization and industry today.*
 - *Threat Prevention: Advanced capabilities like IDPS and malware detection for blocking threats.*
 - *Data Loss Prevention (DLP): Monitoring and controlling sensitive data transmission.*
 - *Encryption and Decryption: Inspecting encrypted traffic without compromising privacy.*
- *Threat Prevention: Proactively identifying and mitigating security threats is essential for maintaining the integrity of network infrastructure.*
 - *Intrusion Detection and Prevention: Monitoring for suspicious or malicious activity.*
 - *Application Control: Granular control over specific applications or services.*
 - *URL Filtering: Controlling access to permitted URLs.*
- *Security compliance: Does your organization have network security requirements based on industry or organization compliance? For example - SAMA (Saudi Arabia Monetary Authority), HIPAA (Health Insurance Portability and Accountability Act), GDPR (General Data Protection Regulation), SWIFT, etc.*

Example:

The standard implementation for [customer] is to use Hub & Spoke networking, leveraging Azure Firewall service in the shared Hub VNet to route and inspect traffic. However, for the purpose of these workloads the applications and databases shall be placed in separate VNets, which will be directly peered. Traffic between application and database VNets does not have to go through the central Hub. This is in order to achieve the lowest possible latency.

Additionally, users who are located on premises, and customers from the internet will have access to the application(s).

DNS for Oracle Database@Azure will be integrated with the existing Azure DNS environment.

3.2.6 Management and Monitoring

Guide:

This subsection helps you capture any requirements for customer management and monitoring needs - e.g. system monitoring, systems management, log analysis, etc.

When you move or start an OCI or OD@Azure project, you have a choice to use the tools you are familiar with (should they support modern application architectures), replace them with OCI or Azure native Observability services, or use a combination to improve your visibility. When contemplating how to proceed, here are some general questions that will guide you:

- *Does the tool manage across hybrid and multi-cloud environments?*
- *What is the cost of integrating the existing tool with OCI?*
- *Is my current monitor tool enabling you to prevent issues versus reacting to them?*
- *Does the tool tell you how much impact there has been on users or just that there was an impact like something is down or unavailable?*

- Does the tool provide the full vision of applications and their infrastructure or just a piece of them or specific technology?

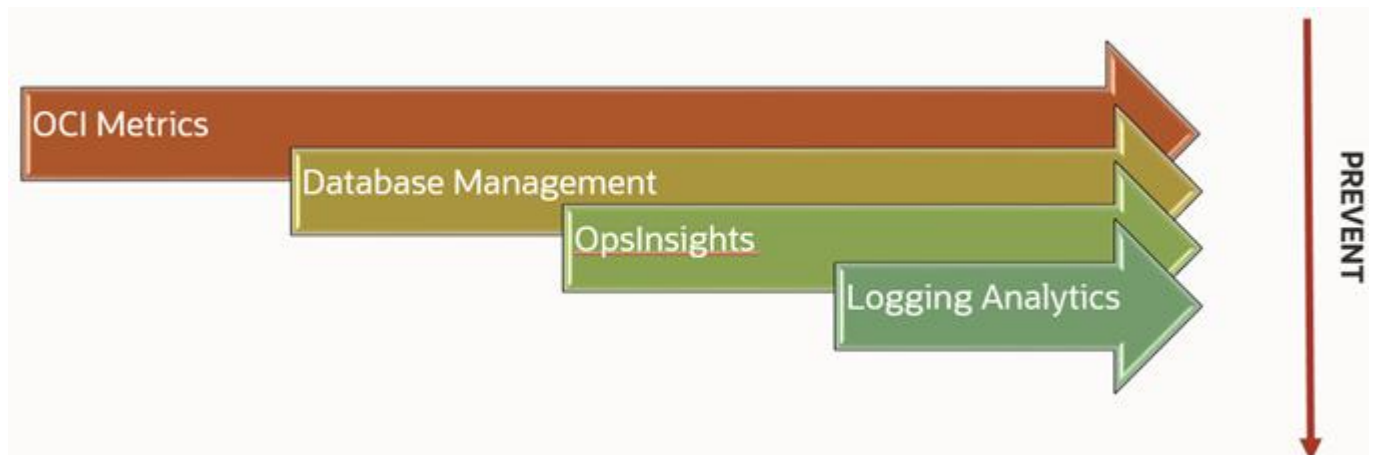
Because OD@Azure runs within an OCI tenancy, Oracle's recommended approach is to leverage native OCI observability services to ensure optimal performance and management. All the logs and monitoring metrics would be available through OCI, with part integration with Azure.

OCI native Observability services for Oracle Database@Azure:

Tool	Service	Tool location for ADB-S	Tool location for ExaDB-D	OCI Subscription	Notes
OCI Metrics	basic monitoring	OCI	OCI/Azure	not required	Available by default
Database Management*	advanced monitoring & admin	OCI	OCI/Azure	required	To be enabled - DB Management Admin features only available in OCI
Ops Insights	Forecast & Capacity planning	OCI	OCI	required	To be enabled
Logging Analytics	Log analysis	OCI	OCI	required	OOTB Parsers/dashboards for Oracle DB

OD@Azure natively exposes logs and maps basic metrics. To enable Administration and advanced analytics capabilities like Resource Forecast, OCI subscription is needed as the relevant services cannot be consumed against Azure MACC. Once the Oracle Universal credit are available it is possible to activate OCI native services as described here. OOTB dashboard and metrics are available on OCI tenant.

*:Oracle Enterprise Manager provides Database Management capabilities for customers who do not have an OCI subscription, offering a comprehensive solution for managing Oracle database.



Observability

Example:

[customer] wants to use Azure Monitor for Metrics, and OCI Database Management, and Splunk for log analysis.

Tool	Task	Target	Location	New	Notes
OCI Database Management	Manage DB Instances + Monitoring and troubleshooting	All targets	OCI	Yes	
Azure Monitor	Basic metric monitoring	All VM Clusters and DBs	Azure	Yes	

Tool	Task	Target	Location	New	Notes
Splunk	Log Analytics	All targets	Splunk	No	

3.2.7 Automation & Integration (IaC)

Guide:

This subsection captures any requirements for integration into the customer's existing automation tools and practices - e.g. terraform, API, SDK requirements. Also, if the customer requires integration into automation platforms e.g. Jenkins, Ansible, GitHub, CD/CI Azure pipelines etc. these should be recorded.

Example:

At the time of this document creation, no Automation requirements have been specified.

[Customer] can leverage Azure and OCI Cloud Native automation tools to provision end to end Exadata Infrastructure, Cluster VM and the database. Both providers are declared and you can use one Terraform recipe to orchestrate Exadata Infrastructure, VM Cluster and the PDB and CDB provisioning and management.

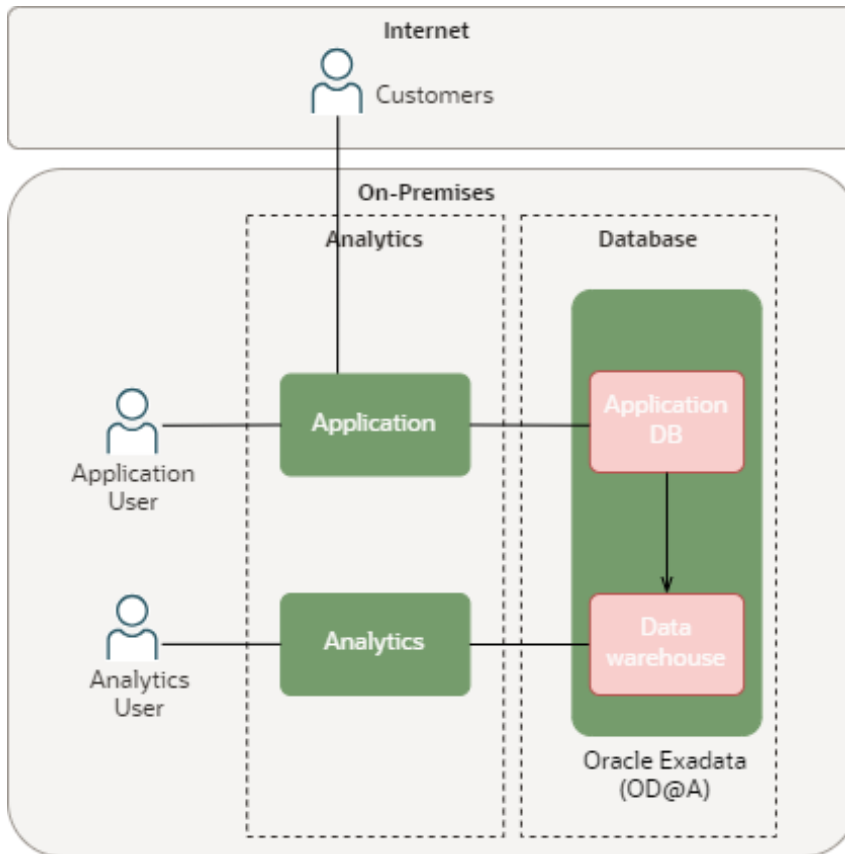
[Official documentation](#) can also be referenced.

3.3 Current State Architecture

Example:

The current implementation of this workload runs fully in [customer]'s own data center. 1 Exadata Machine contains all database workloads, and both the application and the BI environment are running in the same network.

Internet-based customers have access to the self-service part of the application.



Current State Architecture

3.4 Future State Architecture

Guide:

The Future State Architecture can be described in various forms. In the easiest case, we describe a Logical Architecture, possibly with a System Context Diagram. A high-level physical architecture is mandatory as a description of your solution.

This should be the final architecture as part of the pre-sales solution, not an intermediate or draft version

Additional architectures, in the subsections, can be used to describe needs for specific solution.

3.4.1 Solution Best Practices

Oracle highly recommends following best practices to protect your solution:

Azure / OCI Mandatory security best practices:

Oracle Database@Azure is deployed within an Azure framework, and therefore any Azure security best practices apply to the whole solution / workload. Those best practices are not discussed further in this document.

Oracle Database@Azure is also part of an OCI tenancy, and as such OCI Security best practices also apply and are described here:

The safety of the <Customer Name>'s Oracle Cloud Infrastructure (OCI) environment and data is the <Customer Name>'s priority.

The following table of OCI Security Best Practices lists the recommended topics to provide a secure foundation for every OCI implementation. It applies to new and existing tenancies and should be implemented before the Workload defined in this document will be implemented.

Workload-related security requirements and settings like tenancy structure, groups, and permissions are defined in the respective chapters.

Any deviations from these recommendations needed for the scope of this document will be documented in the chapters below. They must be approved by <Customer Name>.

<Customer Name> is responsible for implementing, managing, and maintaining all listed topics.

CATEGORY	TOPIC	DETAILS
User Management	IAM Default Domain	Multi-factor Authentication (MFA) should be enabled and enforced for every non-federated OCI user account. - For configuration details see Managing Multi-Factor Authentication. In addition to enforcing MFA for local users, Adaptive Security will be enabled to track the Risk Score of each user of the Default Domain. - For configuration details see Managing Adaptive Security and Risk Providers.
	OCI Emergency Users	A maximum of three non-federated OCI user accounts should be present with the following requirements: - Username does not match any username in the Customer's Enterprise Identity Management System - Are real humans. - Have a recovery email address that differs from the primary email address. - User capabilities have Local Password enabled only. - Has MFA enabled and enforced (see IAM Default Domain).
	OCI Administrators	Daily business OCI Administrators are managed by the Customer's Enterprise Identity Management System. This system is federated with the IAM Default Domain following these configuration steps: - Federation Setup - User Provisioning - For configuration guidance for major Identity Providers see the OCI IAM Identity Domain tutorials.
	Application Users	Application users like OS users, Database users, or PaaS users are not managed in the IAM Default Domain but either directly or in dedicated identity domains. These identity domains and users are covered in the Workload design. For additional information see Design Guidance for IAM Security Structure .
Cloud Posture Management	OCI Cloud Guard	OCI Cloud Guard will be enabled at the root compartment of the tenancy home region. This way it covers all future extensions, like new regions or new compartments, of your tenancy automatically. It will use the Oracle Managed Detector and Responder recipes at the beginning and can be customized by the Customer to fulfill the Customer's security requirements. - For configuration details see Getting Started with Cloud Guard. Customization of the Cloud Guard Detector and Responder recipes to fit the Customer's requirements is highly recommended. This step requires thorough planning and decisions to make. - For configuration details see Customizing Cloud Guard Configuration
	OCI Vulnerability Scanning Service	In addition to OCI Cloud Guard, the OCI Vulnerability Scanning Service will be enabled at the root compartment in the home region. This service provides vulnerability scanning of all Compute instances once they are created. For configuration details see Vulnerability Scanning.
Monitoring	SIEM	Continuous monitoring of OCI resources is key for maintaining the required security level (see

CATEGORY	TOPIC	DETAILS
	Integration	Regulations and Compliance for specific requirements). See Design Guidance for SIEM Integration to implement integration with the existing SIEM system.
Additional Services	Budget Control	OCI Budget Control provides an easy-to-use and quick notification on changes in the tenancy's budget consumption. It will be configured to quickly identify unexpected usage of the tenancy. For configuration details see Managing Budgets

OCI Landing Zones:

When OD@Azure is implemented, it is always linked to an OCI tenancy.

An OCI Landing Zone sets the foundations for a secure tenancy, providing design best practices and operational control over OCI resources. A Landing Zone also simplifies the onboarding of workloads and teams, with clear patterns for network isolation and segregation of duties in the organization, which sets the cloud operating model for day-to-day operations.

Oracle highly recommends the use of an OCI Landing Zone for any deployment. Use these [guidelines](#) to set up your OCI Landing Zone, including design considerations, approaches, and solutions to use.

Note that all workloads in a tenancy should sit on top of a Landing Zone, meaning that the workload architecture defined in the next section can be subject to adjustments (e.g., network structure) towards the landing zone model, along with other future workloads.

Note also that there are 2 types of OD@Azure environment. Those environments where the OCI tenancy is used for other services should design an OCI Landing Zone according to the normal considerations for those workloads. For those environments where the OCI tenancy is only used to support the Oracle Database @ Azure, a minimal landing zone can be used. In that case we still need to consider items such as:

- Compartments (automatically created by the OD@Azure service)
- IAM (groups and policies created as part of the OD@Azure service)
- Security posture (are services like Cloud Guard, Security Zones, Vulnerability Scanning needed on OCI)
- Network Structure (VCNs automatically created by the OD@Azure service to mirror Azure VNets)
- Network Security
- Network Connectivity
- DNS

Naming conventions:

For OD@Azure workloads without any additional OCI components, it is acceptable to refer to existing Azure naming conventions, rather than describing Azure naming conventions here.

A naming convention is an important part of any deployment to ensure consistency, governance, and security within your tenancy. Find [here](#) Oracle's recommended best practices.

3.4.2 Logical Architecture

Guide:

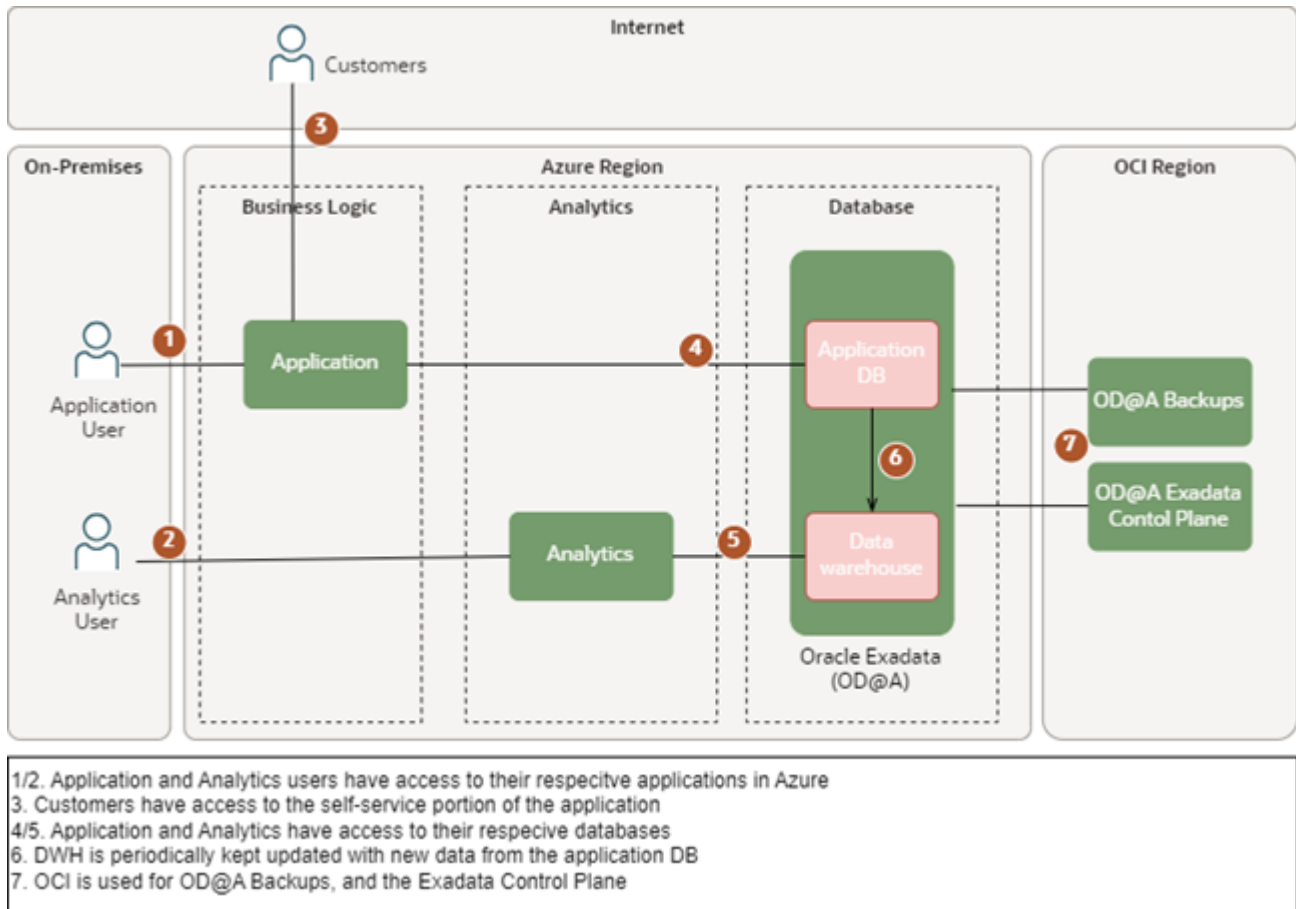
Provide a high-level logical Oracle solution for the complete solution. Indicate Oracle products as abstract groups, and not as physical detailed instances. Create an architecture diagram following the latest notation and describe the solution.

It is expected to provide a full Multicloud architecture in this section, including all Azure elements which need to have access to OD@Azure.

To implement a solution the Physical Architecture is needed in the next chapter. The physical notation can show individual components with physical attributes such as IP addresses, hostnames, or sizes.

The Oracle Cloud Notation, OCI Architecture Diagram Toolkits

Example:



Logical Architecture

Additional examples:

Database consolidation with multiple service levels

Based on requirements for data residency and local application performance this solution delivers Database as a Service (DBaaS) in Azure data centers using ExaDB-D. Considering Oracle Maximum Availability Architecture (MAA) multiple service levels have been defined in the logical solution architecture below.

The logical architectures have multiple service levels for Production Database Service A (Standby in same Azure region), and Production Database service B (Standby across Azure regions).

Production Database Service A

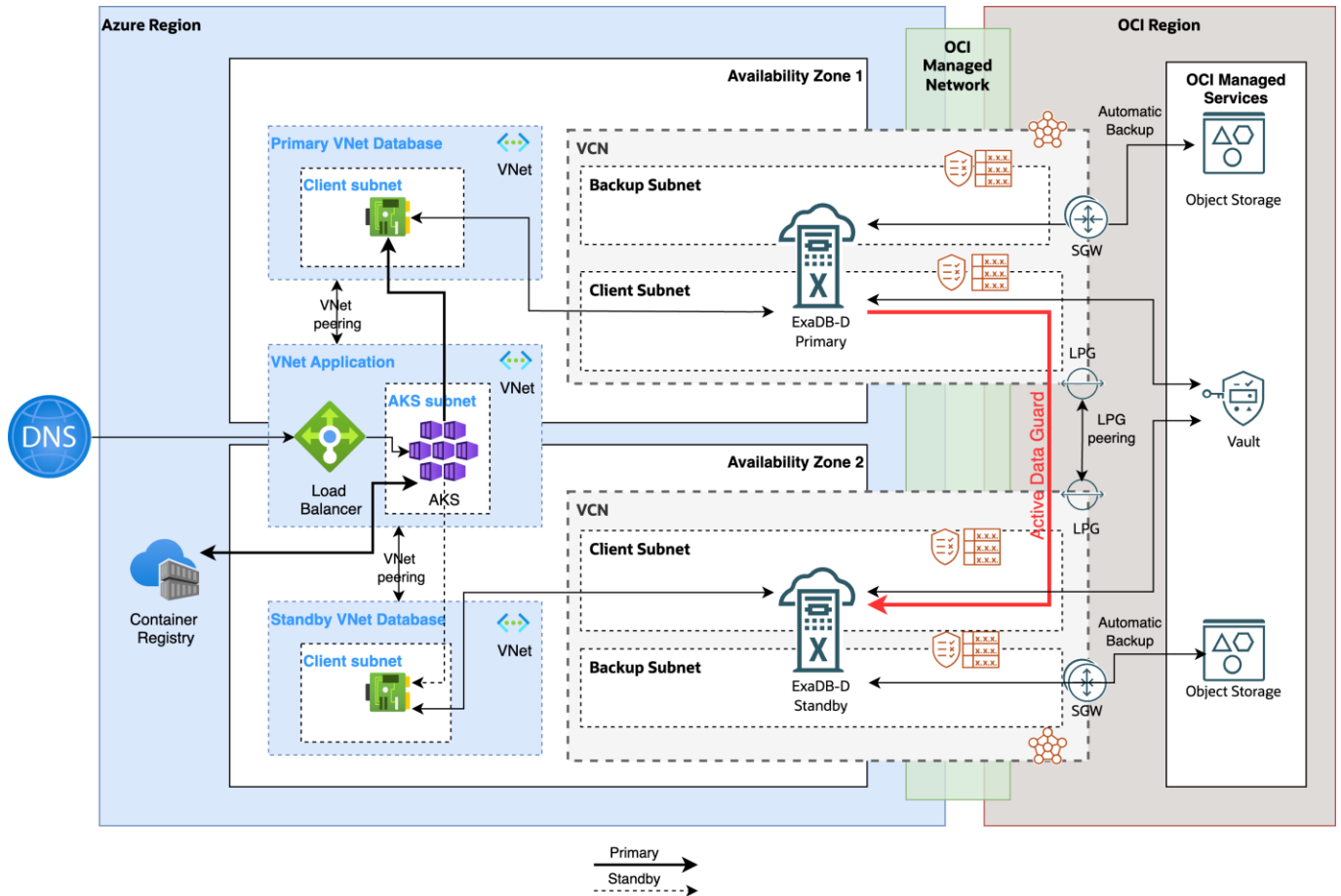
Using MAA Reference Architecture 'Gold' for Production Database Service A which has High Availability with Oracle RAC, Disaster Recovery with Oracle Data Guard and Data Protection with copies in both regions with Oracle Zero Data Loss Recovery Service.

Non-Production Database Service B

Using MAA Reference Architecture 'Silver' for B and C since they are identical but split across two data centers for efficient use of available resources. Non-Production Database Service B and C has High Availability with Oracle RAC and Data Protection with copies in both data centers with Oracle Zero Data Loss Recovery Appliance (ZDLRA) and ZDLRA Replication.

Additional information on MAA Reference Architectures can be found here: [Oracle MAA For Oracle Database@Azure](#)

Service Level with High Availability, Disaster Recovery and Data Protection within region



Details of Database Service A based on MAA Reference Architecture 'Gold' in same region



Details of Database Service A based on MAA Reference Architecture 'Gold' across regions

3.4.3 Physical Architecture

Guide:

The solution architecture is typically described in a physical form. This should include all solution components. You do not have to provide solution build or deployment details such as IP addresses.

It is expected to provide a full Multicloud architecture in this section, including all Azure elements which need to have access to OD@Azure.

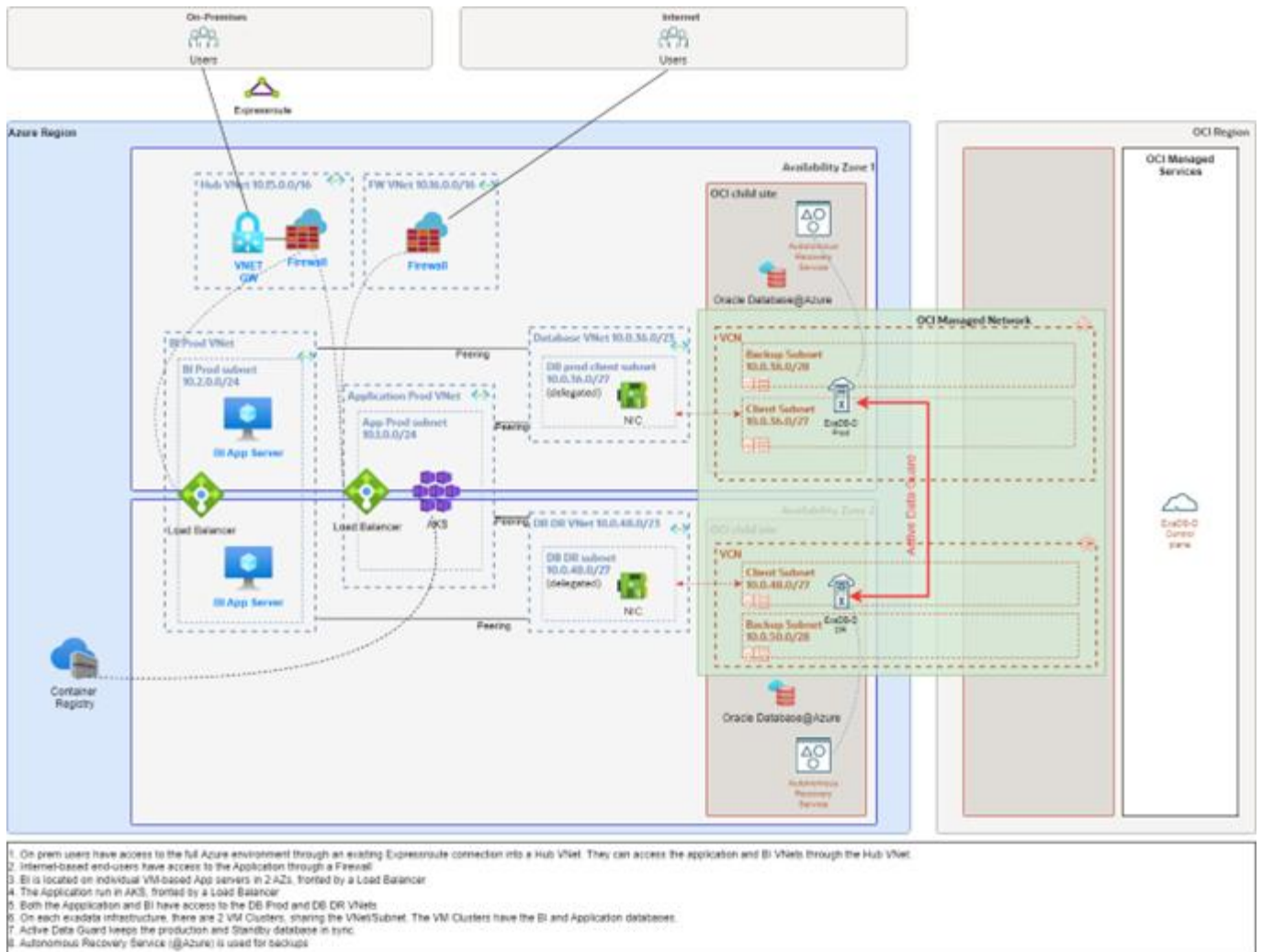
Please describe the solution with an architecture image plus a written text. If you have certain specifics you like to explain, you can also use the Solution Consideration chapter to describe the details there.

The Oracle Cloud Notation, OCI Architecture Diagram Toolkits

Reference:

StarterPacks (use the search)

Example:



Future State Deployment Diagram

In this example only 1 VM cluster is used. If there are multiple workloads across multiple VM clusters, then additionally notes can be made on VM Clusters fulfilling the following requirements:

- Network segmentation due to security: Each VM Cluster can be placed in its own Azure VNet.
- Workload isolation: Each VM Cluster have its own dedicated resources in CPU, Memory and Storage Capacity.
- Application / User separation: Each VM Cluster can have its own users and groups as defined in the Identity Service.
- Required Service Level Objectives (for example MAA Gold, Silver): Each VM Cluster is designed against a given MAA Reference Architecture.

High Availability will be granted with Oracle RAC features.

Information on Oracle RAC Best Practices for DB 19c is here: [Overview of Oracle RAC and Clusterware Best Practices](#).

Additional information on RAC can be found in the MAA Reference Architectures here: [Oracle MAA Reference Architectures](#).

Disaster Recovery capabilities will be granted with Oracle Data Guard.

Information on Oracle Data Guard Best Practices for DB 19c is here: [Oracle Data Guard Best Practices](#).

Additional information on Data Guard can be found in the MAA Reference Architectures here: [Oracle MAA Reference Architectures](#).

In case of unplanned/planned downtime of the production site, an automated or manual mechanism will allow failover/switchover to the other production platform.

Oracle Transparent Data Encryption (TDE) will be used. This is mandatory with ExaDB-D@Azure

The Control Plane of this ExaDB-D@Azure is in the local OCI Public Cloud region. The control plane is a sophisticated cloud software suite for order management, subscription billing, identity management, UI and REST API services, network connectivity, security controls, infrastructure and lifecycle management. Oracle Identity and Access Management (optionally federated with Azure EntraID), lets you control who has access to your Exadata resources. One can control what type of access a group of users have and to which specific resources.

For data protection all VM Clusters on the same Exadata Infrastructure share the same Backup Subnet.

3.4.4 Key Solution Proposal Components

Example:

Our proposal is based on the following main technical components:

3.4.4.1 Exadata Dedicated @ Azure (ExaDB-D@Azure)

Oracle ExaDB-D@Azure brings the performance, automation, and economics of Exadata Database Service and Autonomous Database fully managed by Oracle to data centres in Azure. This is the easiest way for [customer] to start using cloud database resources within Azure while continuing their deployment of Azure. ExaDB-D incorporates unique optimizations that enable databases to run faster with less management and at lower costs, to better leverage [customer]'s data.

Value Proposition

Oracle's value proposition to support [customer]'s needs is built around the Oracle ExaDB-D cloud platform:

Resilience and Availability

- Modern, highly available, and secure data architecture by design
- Simplified patch maintenance and proactive threat management
- Through ExaDB-D@Azure, these services can be used inside an Azure datacenter, natively part of the Azure network.

Performance, agility, and flexibility

- Low latency to keep data as close as possible to applications and choose the timeline for modernizing these databases as needed
- Ability to scale the platform as needs change without increasing complexity
- Maximum availability architecture (e.g. PRA with fast RTO and RPO, service continuity, etc.)
- Obsolescence and risks
- Simplify and standardize to enhance security
- Automation to achieve more predictable service levels and address a wide range of security vulnerabilities Security
- Encryption of all data at rest and in transit.

Speed of implementation / Time to Market

- Time to Market optimization is made possible thanks to the comprehensive 'Infrastructure as Code' approach, which allows all environments to be managed through a DevOps methodology to ensure repeatability.
- Implementation of applications directly and without the need for infrastructure teams, greatly accelerating their implementation and simplifying their maintenance.

3.4.5 Solution Considerations

Guide:

Describe certain aspects of your solution in detail. What are the security, resilience, networking, and operations decisions you have taken that are important for your customer?

Typical key considerations are:

- *High Availability and Disaster Recovery*
- *Security*
- *Networking*
- *Manageability and Observability*

Feel free to describe other aspects and important requirements for your customer, and how your solution addresses them in this chapter.

Security:

Generic security guidelines are in the [Annex](#)

Note that Key Management is always something that needs to be thought of clearly for OD@Azure Exadata deployments. If customer-managed keys are required, then Oracle Key Vault is usually the default correct choice. However, if for example keys need to be held in hardware, then OCI Vault should be used.

An overview of the pros and cons of the various options can be found at: [Overview of key management solutions for OD@Azure](#)

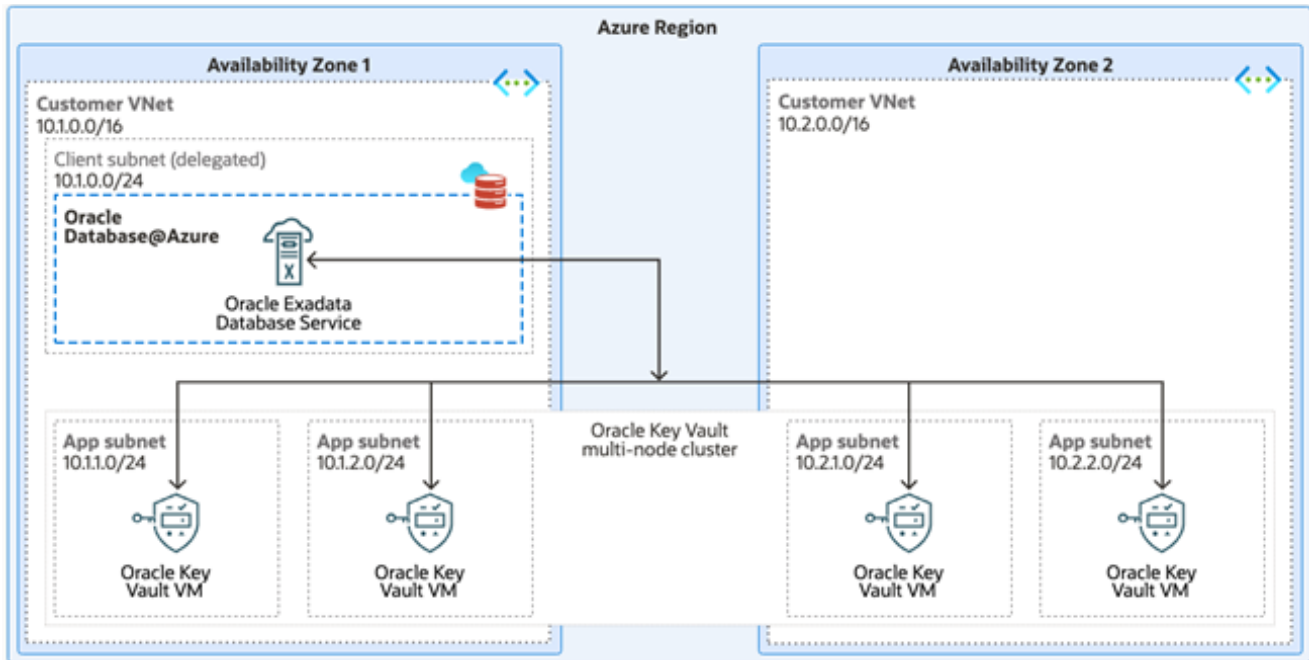
Example:

Key management options for Oracle Database@Azure:

Option	Service		Info
Oracle Managed keys	File based		Default in Exadata Cloud Infrastructure
Customer managed keys	OCI Vault (service in OCI, UCM credits needed)	Virtual Vault Multitenant partition	HSM Keys - A master encryption key protected by an HSM is stored on an HSM and cannot be exported from the HSM. All cryptographic operations involving the key also happen on the HSM. These keys are FIPS Level 3 compliant. Software Keys - A master encryption key protected by software is stored on a server and can be exported from the server to perform cryptographic operations on the client instead of on the server. While at rest, the software-protected key is encrypted by a root key on the HSM. These keys are FIPS Level 1 compliant.
		Private Vault: Own partition in HSM. Better and consistent transactions per second for cryptographic operations. Cross Region Replication. Cross Region Back/Restore of Keys	HSM Keys - A master encryption key protected by an HSM is stored on an HSM and cannot be exported from the HSM. All cryptographic operations involving the key also happen on the HSM. These keys are FIPS Level 3 compliant. Software Keys - A master encryption key protected by software is stored on a server and can be exported from the server to perform cryptographic operations on the client instead of on the server. While at rest, the software-protected key is encrypted by a root key on the HSM. These keys are FIPS Level 1 compliant.
	Azure Key Vault	Azure Key Vault Standard	Azure Key Vault Standard Tier is a FIPS 140-2 Level 1 validated multitenant cloud key management service. Keys stored in Azure Key Vault are software-protected. Also review Azure Documentation
		Azure Key Vault Premium	Azure Key Vault Premium uses hardware security module(HSM)-protected keys. Also review Azure Documentation
		Azure Key Vault Managed HSM	Azure Key Vault Managed HSM (Hardware Security Module) is a fully managed, highly available, single-tenant, standards-compliant cloud service using FIPS 140-3 Level 3 validated HSMs. Also review Azure Documentation
	Oracle Key Vault (license needed)		Software on VM, manual installation. Located on Azure, "local" to the database.

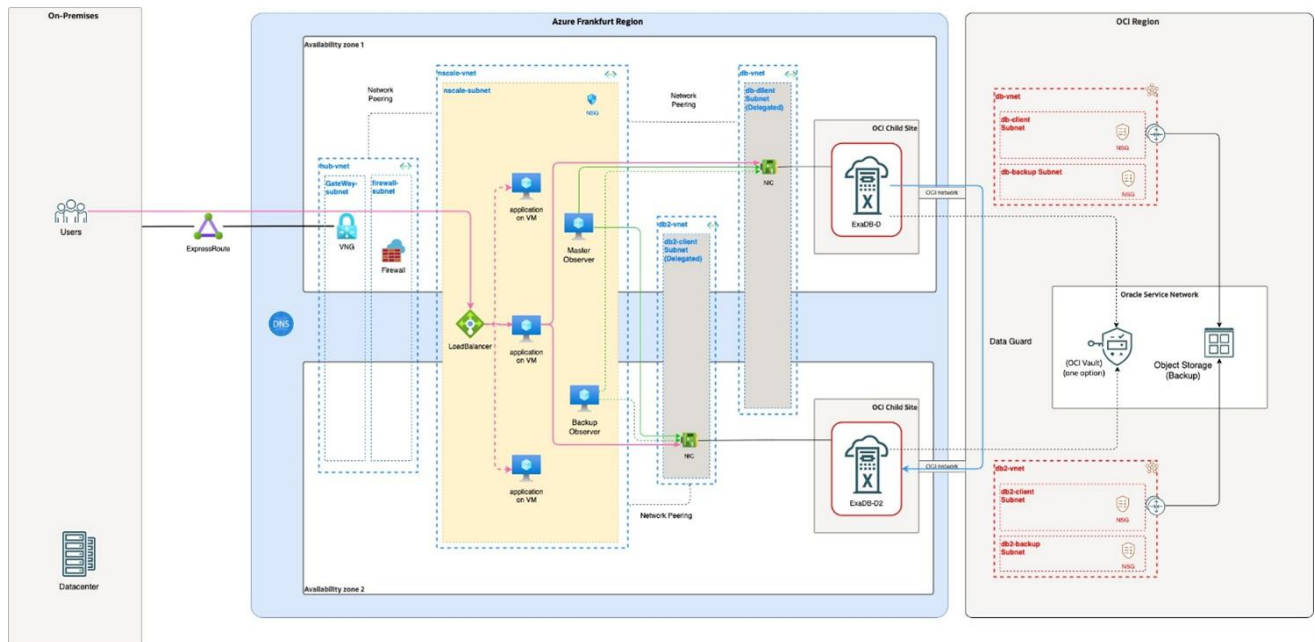
Links for [Azure Key Vault Standard & Premium Documentation](#) and [Azure Key Vault Managed HSM Documentation](#).

OD@Azure with OKV on Azure VM:



ODAA with OKV on Azure VM

OD@Azure with OCI Vault:



ODAA with OCI Vault

Networking:

Guide:

Please describe your solution from a networking design point of view.

Example:

Hub and Spoke with 1 VM cluster in a single region:

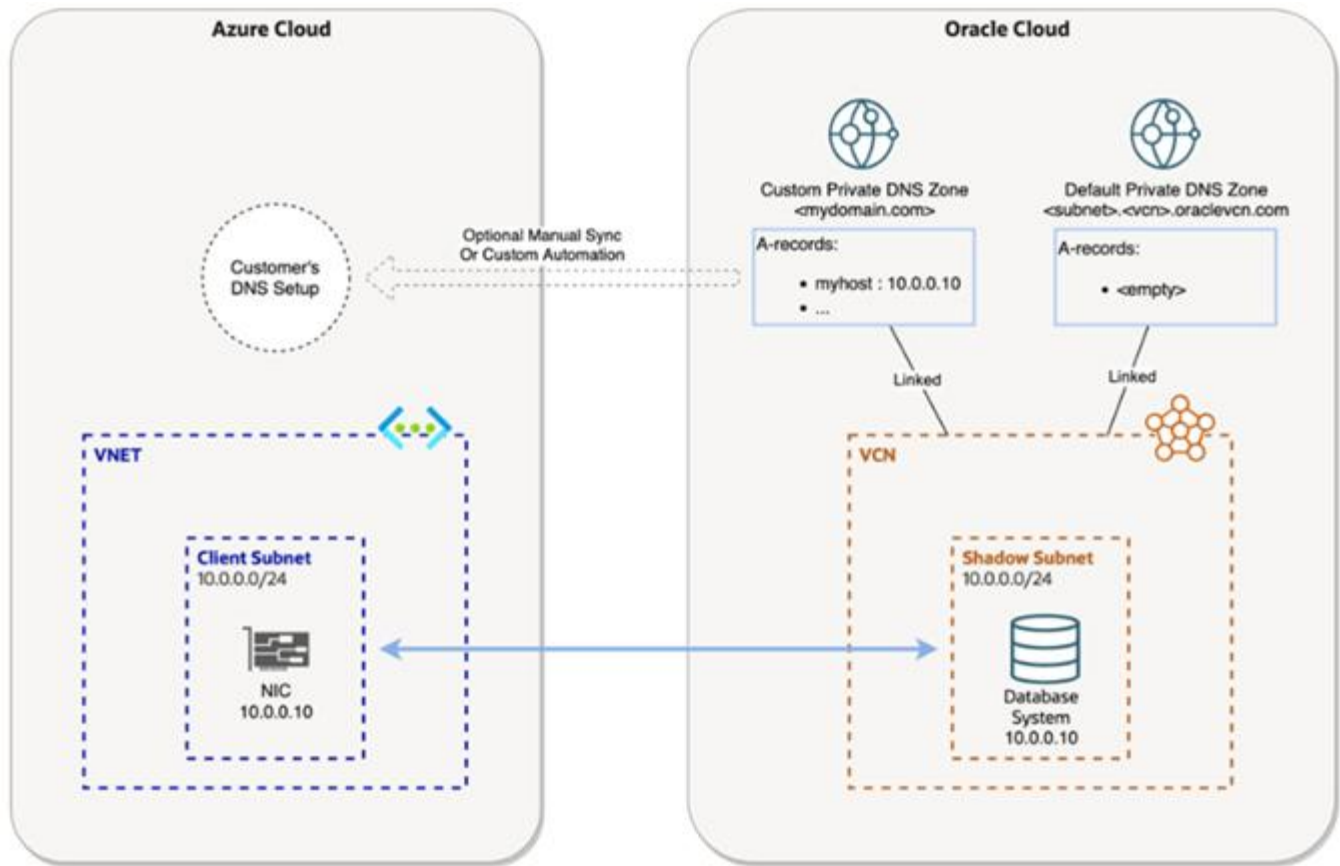


Hub and Spoke 1 region

Multiple application VNets connect to a Hub VNet, and from are routed to the Database VNet. Routing within the Hub VNet is done through the Azure Firewall service.

The OD@Azure service replicates the VNet setup to create a shadow-VCN in OCI. A Backup Subnet is created in this VCN, through which the backup traffic is routed to the OCI Backup location.

Additionally, describe any requirements with regards to DNS:



DNS

In order to use a Custom DNS solution, an OCI private DNS zone needs to be created with the right name and records.

During VM Cluster creation, in the 'Use private DNS service' checkbox can be selected, which allows the entry of the relevant OCI DNS Zone.

Reliability and Resilience:

Proposed HA/MAA design and details

Guide:

- *Resilience on OCI*
- *Workload Related Content*
- *What happens when site 1 is down?*
- *What happens when XYZ happens?*
- *How is HA achieved?*
- *What Backup/Recovery options are chosen?*

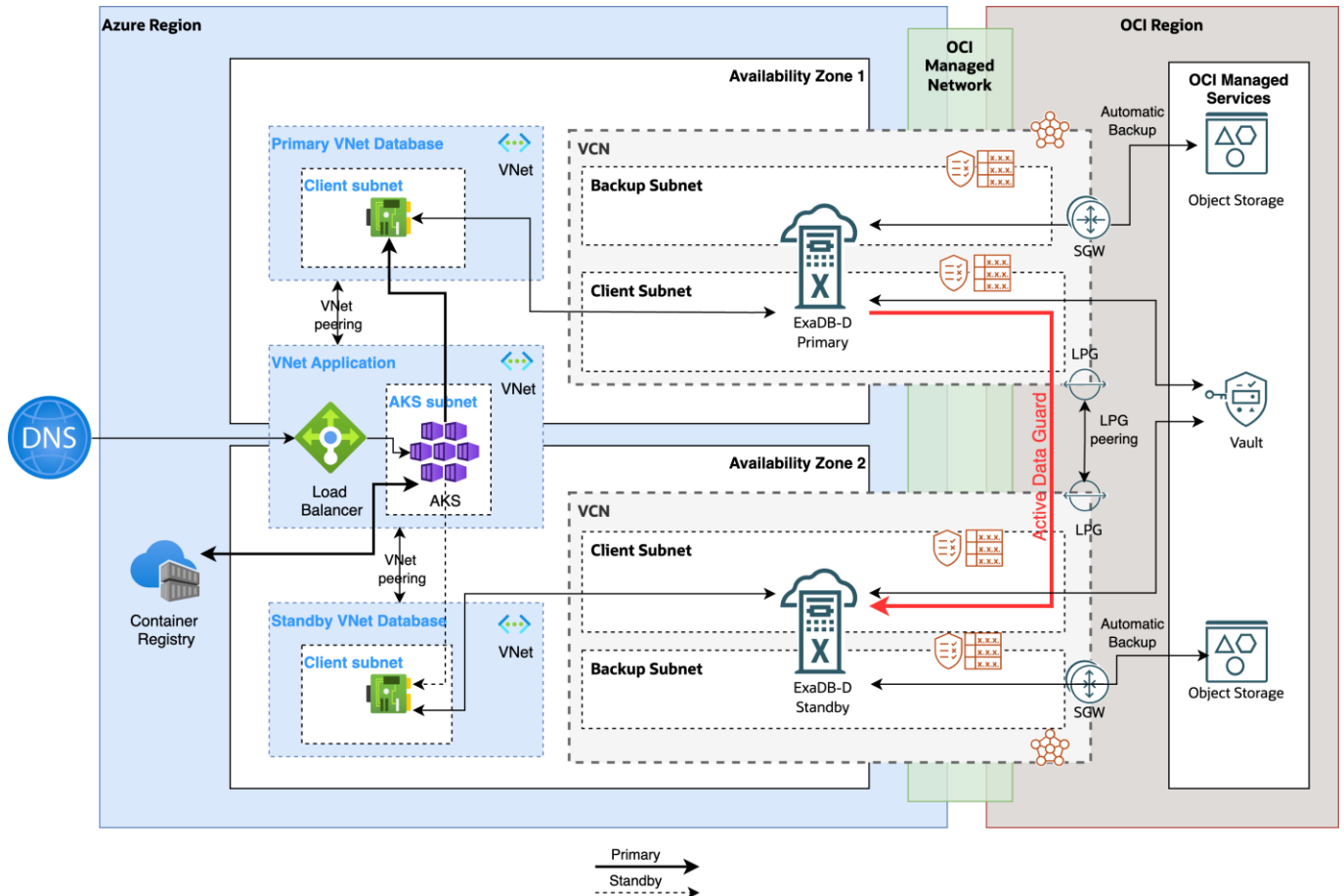
Example:

Customer requires to adhere to Oracle MAA principles within the same region.

The following has been decided:

- Production Databases:

- Database Service A
- MAA Gold with 2-node dedicated VM-cluster in AD1 and 2-node dedicated VM-Cluster in AD2
- Full RAC
- DataGuard with AD1 as primary and AD2 as standby
- Backups will be done from the local instance to the local backup environment
- Fast-Start Failover to automate the database failover and reduce the recovery time without performing any manual steps. RPO within seconds



ADG single region

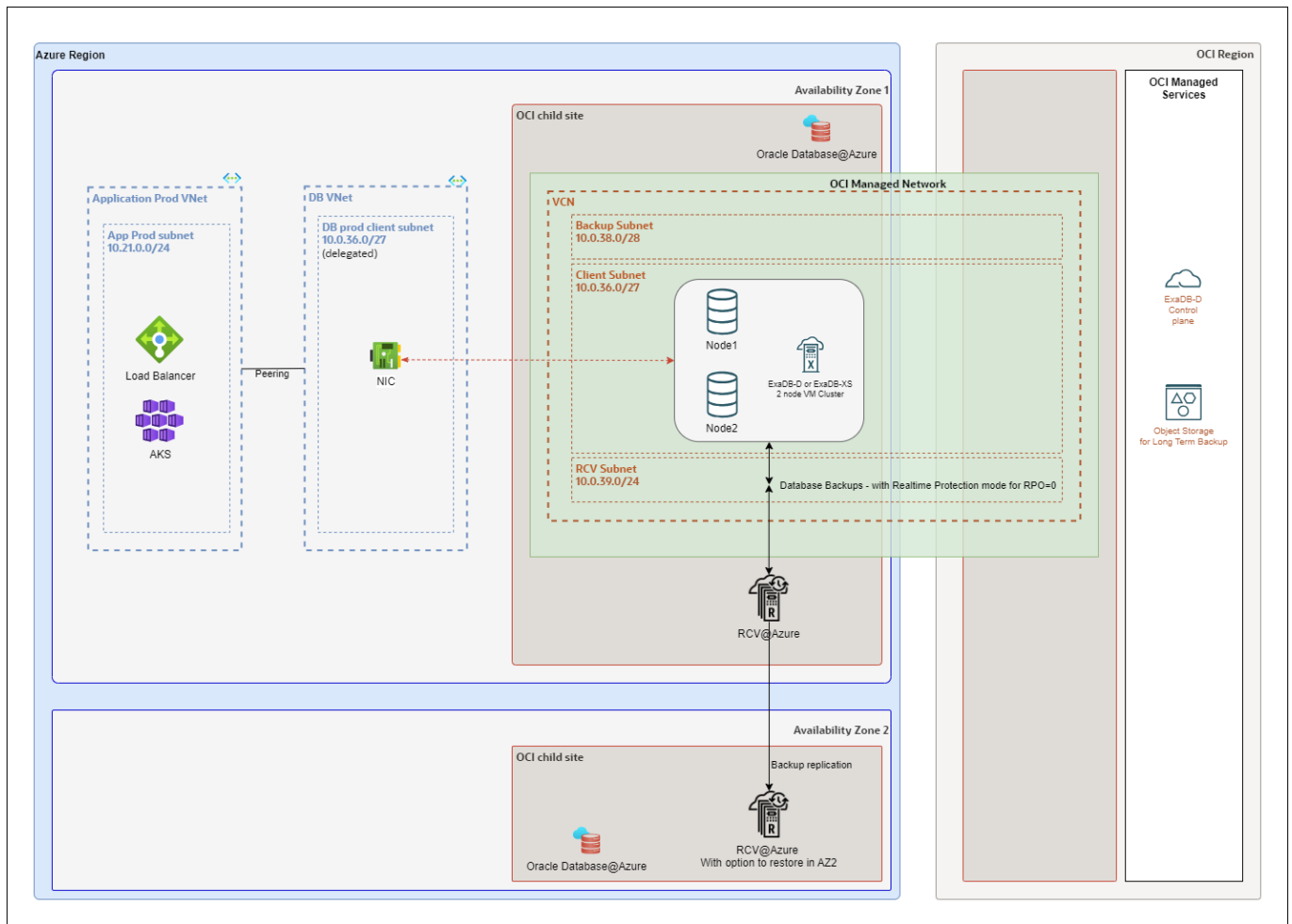
Proposed DR design and details

Guide:

- What Backup/Recovery options are chosen?
- Oracle Autonomous Recovery Service should always be positioned first, either @Azure or @OCI
- As an alternative OCI Object Storage can be used.
- If a customer really wants it, then Azure Blob Storage (or any other target) can be used with manual backups, this should not be recommended.

Example 1:

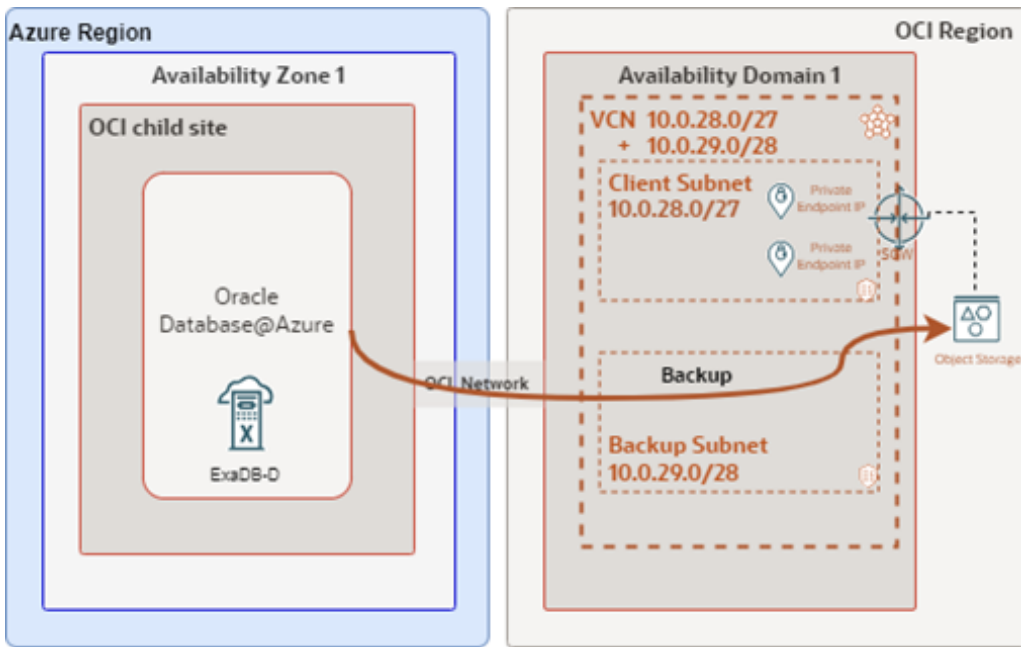
Exadata in 1 AZ with Autonomous Recovery Service:



RCV

Example 2:

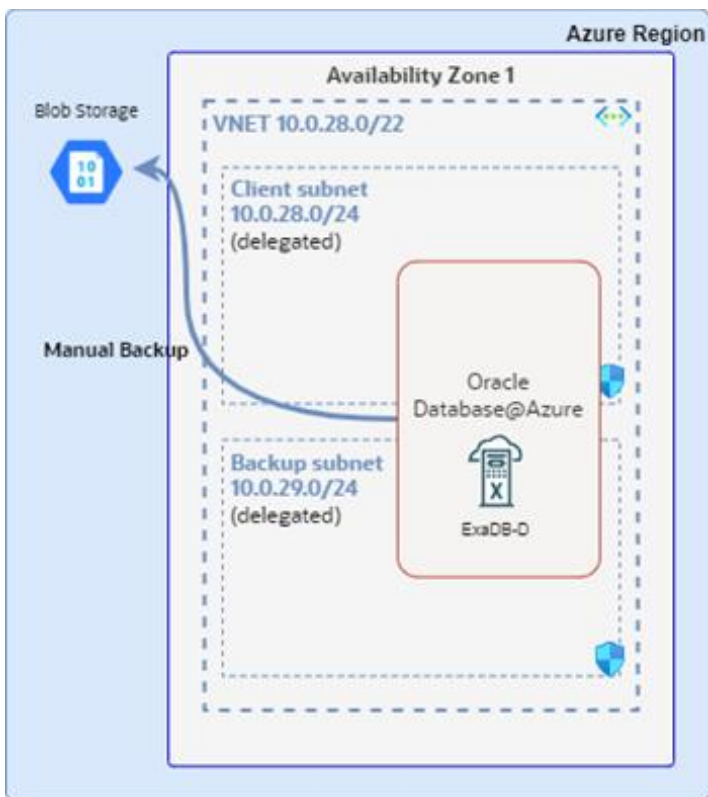
Automatic Backup to OCI Object Storage:



backup to Object Storage

Example 3:

Manual Backup to Azure Blob storage:



backup to azure Blob

Monitoring:

Guide:

Describe your solution for monitoring options

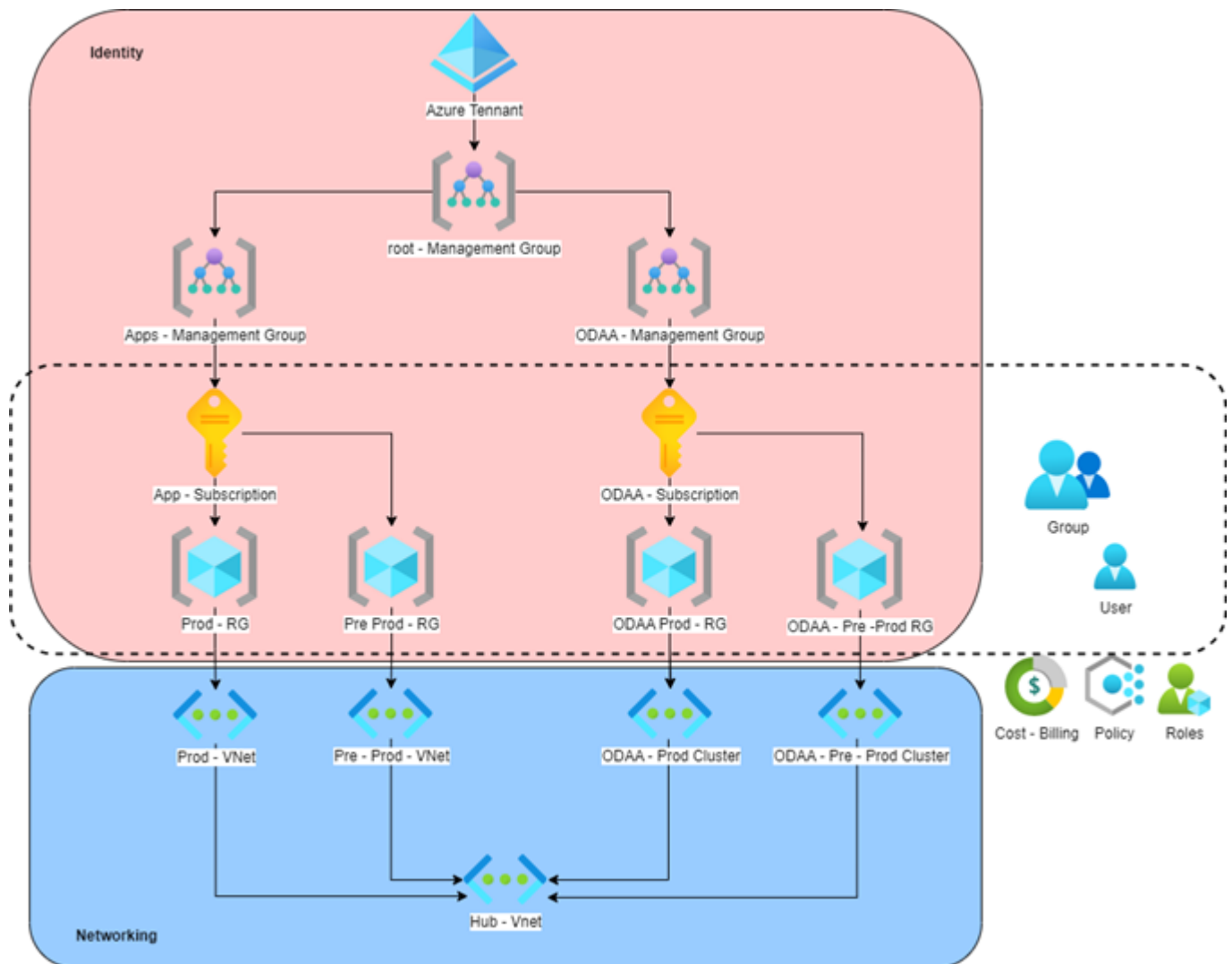
- *Azure Log Analytics*
- *Azure Application Insight*
- *Oracle Enterprise Manager*
- *OCI Observability*
- *3rd party tools (Splunk etc)*

Governance:

Guide:

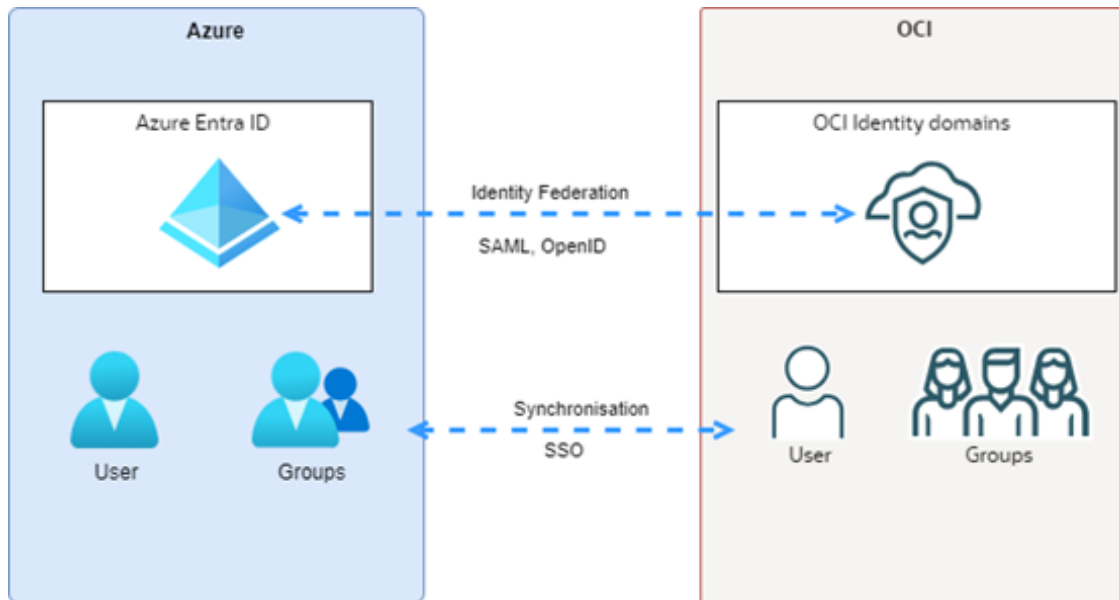
Describe the Azure governance model (subscription, resource group)

OD@Azure IAM:



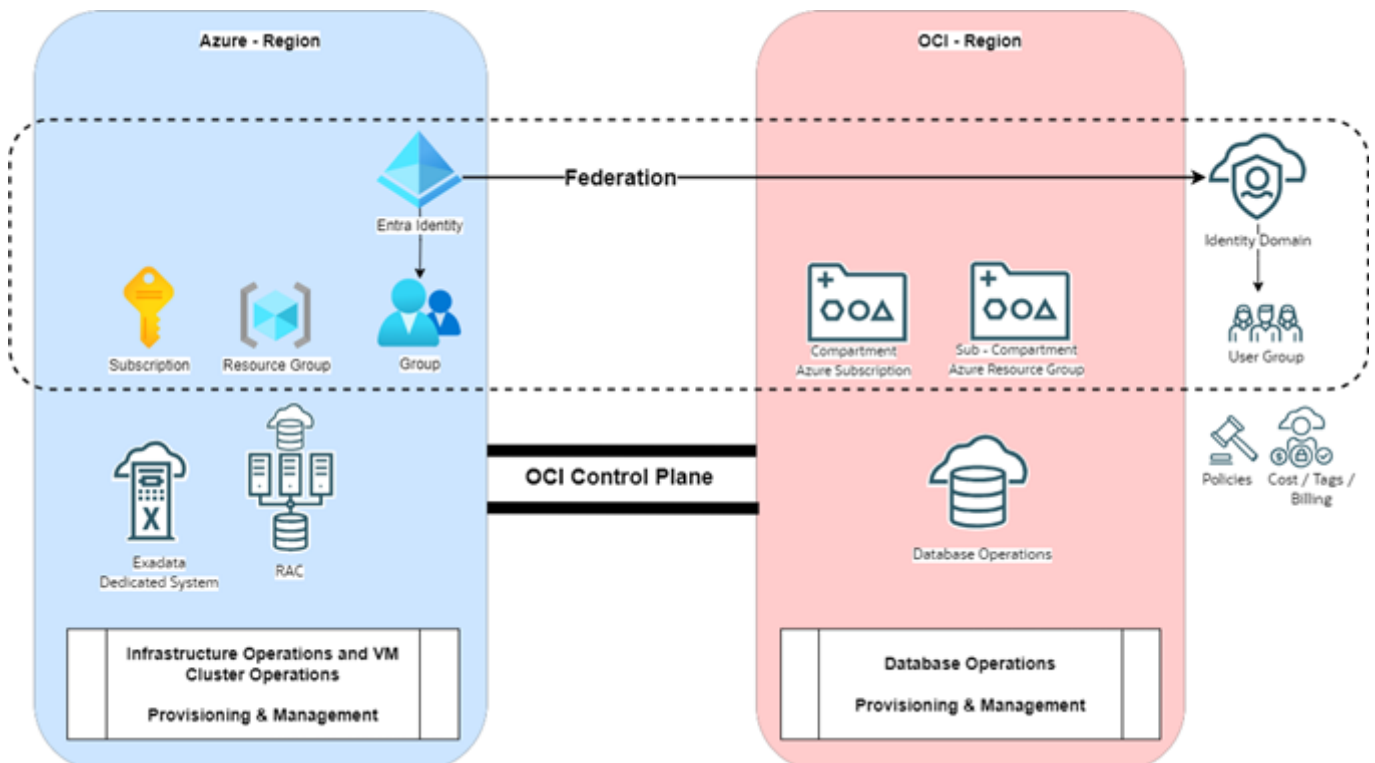
IAM

OD@Azure Identity Federation:



Federation

OD@Azure Operations:



ODAA Operations

Oracle Database@Azure Operating Model:

Guide:

This section captures requirements on how the system will be managed after implementation and migration.*

Reference can be made to the normal 'Explanation of Cloud Operations Service' for ExaDB-D in [MOS Note 2875973.1](#). The split between Oracle and customer responsibility is the same for ExaDB-D running within Azure as it is for ExaDB-D in OCI.

Example:

[customer]'s full Azure cloud estate, including Oracle Database @ Azure is managed fully by [partner].

For the management of the Exadata Database Service on Dedicated Infrastructure there is a split between Oracle responsibility and Customer/Partner responsibility, as described in 'Explanation of Cloud Operations Service' for ExaDB-D in [MOS Note 2875973.1](#).

In short Oracle owns and is responsible for managing the Infrastructure layer, and the customer/partner manages the layers on top of that (VMs, Databases, including access to these).

For an overview of responsibilities between Azure, Oracle and the end customer, see section 'Roles and Responsibilities for OD@Azure' of the document [Exadata Database Service on Dedicated Infrastructure Security Controls](#).

For incident management, Oracle Cloud Support is the first line of support for all Oracle Database@Azure and questions. Detailed information on Oracle and Azure support responsibilities are available [in the documentation](#).

3.5 Sizing and Bill of Materials

Guide:

Estimate and size the physically needed resources of the solution. The information can be collected and is based upon previously gathered capacities, business user numbers, integration points, or translated existing on-premises resources. The sizing is possibly done with or even without a Physical Architecture. It is okay to make assumptions and to clearly state them!

Clarify with sales your assumptions and your sizing. Get your sales to finalize the BoM with discounts or other sales calculations. Review the final BoM and ensure the sales are using the correct product SKUs / Part Number.

Even if the BoM and sizing were done with the help of Excel between the different teams, ensure that this chapter includes or links to the final BoM as well.

3.5.1 Bill of Materials scope

This section covers only the Oracle Database@Azure elements, and any additional OCI elements. It does not contain any Azure services or components.

3.5.2 Collected Data for Sizing

Guide:

With OCCA Oracle Cloud Capacity Analytics the data collection comes from Oracle Enterprise Manager, AWR (Automatic Workload Repository) and/or is manually added.

Customer data is gathered with a script provided by Oracle with clear instructions on how to execute. It's customer responsibility to run the extract script and optionally obfuscate the data, and then upload the output to Oracle using secure methods.

The output needs a first validation, where any anomalies in the data collection, such as errors, missing data, unrealistic outliers/deviations or unexpected growth can be identified.

This data is then grouped in Cohorts which are in the context of sizing grouping of databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:

- *Network segmentation due to security*

- *Workload isolation*
- *Application / User separation*
- *Required Service levels (for example MAA Gold, Silver, Bronze)*
- *Size of the workload*
- *Physical Location*

data collection

Document the scope, manual additions, corrections and/or assumptions made related to the anomalies.

Example:

The scope that has been studied considers all the Oracle databases within the customer. This scope includes both Oracle Databases hosted on on-premise Exadata Infrastructures and Oracle Databases on legacy infrastructure (VMWare).

To complete the study, Oracle provided customer with scripts to capture the activity of the Oracle databases included in this scope. However, the activity of a number of databases could not be recovered. For these databases, we relied on assumptions shared by customer.



data collection

The data collection showed vCPU growth from August 2024 to September 2024 was higher than expected and because of that checked with and confirmed by customer. This will result in adjusted sizing values for OCPU.

3.5.3 Defined VM Clusters for Sizing

Guide:

Cohorts in the context of sizing are grouping databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:

- *Network segmentation due to security*
- *Workload isolation*
- *Application / User separation*

- *Required Service levels (for example MAA Gold, Silver, Bronze)*
- *Size of the workload*
- *Physical Location*

Document the VM Clusters that has been agreed with the customer and any assumptions made. If you have a more complex VM Cluster layout with VM Cluster Node Subsetting, then consider visualizing it.

Example:

Based on the customer's different requirements for Production and non-Production environments which include different service levels as well as separate Client network segments for the environments, the following Cohorts (VM Clusters) have been defined.

Each VM cluster runs a single version of the Oracle Grid infrastructure (GI) to be chosen by the customer among the most recent. Each cluster can run different versions of Oracle Homes. Each independent VM cluster has OCPU assigned during the deploy phase of the VM cluster, so, we can assign different number of OCPU to different VM clusters and environments.

VM Clusters to be deployed are listed in the following tables together with the criteria that define why separate VM Clusters are deployed.

VM Clusters	Environment	VNet	MAA Service Level
Production A Primary	Production	DBVNetProd	Gold
Non-Production B	Test	DBVNetNonProd	Silver
Miscellaneous D	Test	DBVNetMisc	Silver

An ExaDB-D@OCI X9M Quarter Rack (with two DB Servers) can host up to X independent VM Clusters, the maximum number of VM Clusters is Y for X9M systems with more than 2 database servers. The maximum number of VMs per database server is Z regardless of the number of database servers in the system.

Documentation on [Managing VM Clusters in ExaDB-D@Azure](#) is also available.

3.5.4 Sizing Analysis

Guide:

Each VM Cluster is sized independently using Time-Aligned Peak Sizing with adjustments made for historical deviations and finally all VM Clusters are combined per ExaDB-D rack.

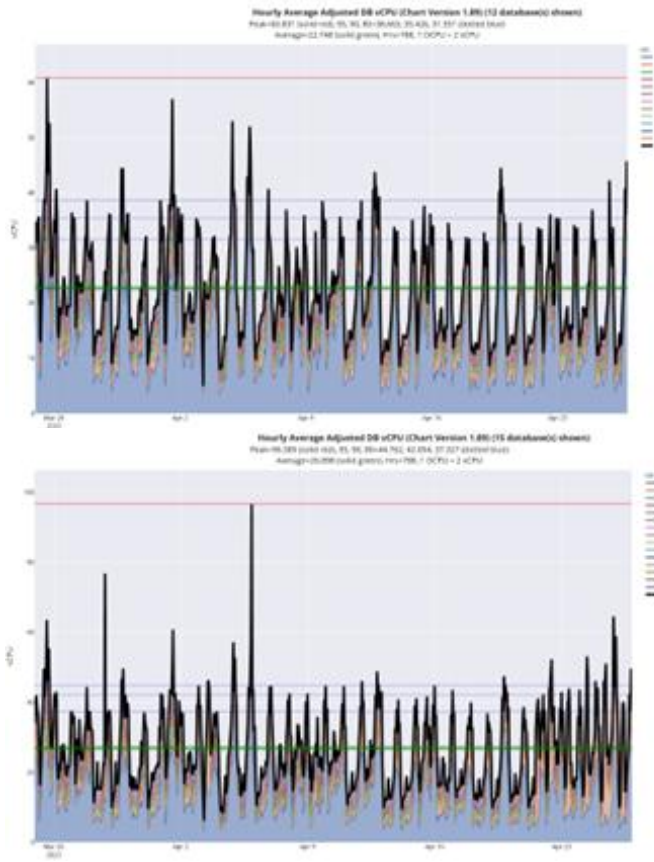
Values of interest include:

- *OCPU Peak*
- *OCPU 95th Percentile*
- *DB Memory*
- *Allocated and Uses Storage*
- *IOPS Read and Write*

Document the summary from the Sizing Summary for customer reference.

Example:

These are the time aligned plots for the two production VM Clusters of Database Service A (Primary + Standby).



3.5.5 Sizing Assumptions

Guide:

Document the customer requirements that is included in the sizing as well as assumptions made for the sizing. This information is provided as part of the Sizing Summary document.

Example:

The following customer requirements and general assumptions have been used in the sizing.

- All analyses are based on previous resource consumption and does not necessarily predict the future workload behavior.
- The defined cohorts (VM Clusters) and databases can differ from the current deployment.
- The sizing computations are based on the hourly average resource consumption of databases in each cohort.
- Cohort metrics are aligned by time and then the sum of peaks is computed for each hour.
- Hourly data is adjusted based on peaks found in the daily data over the past year (valid for EM data case).
- Using in-memory or other database options may reduce/increase the required processing power.
- Using different execution plans will influence the required processing power.
- All Prod DBs will have DR. 5% YoY growth was considered for a period of 4 years (corresponding system shape was selected).
- 10% additional overhead has been considered for converting standalone DBs to RAC.
- For each VM Cluster, an additional 4 OCPUs were added for OS+GI. Metrics for DR DBs were considered as 20% of the corresponding Primary DBs.

3.5.6 Proposed Hardware Shapes

Guide:

Document the proposed hardware shape and any additions, corrections or customer requested changes made.

Example:

The sizing shows that ExaDB-D@Azure X9M Quarter Rack will fulfill requirements, but due to heavy use of CLOB in some Production Databases the customer requests to add an additional ExaDB-D@Azure X9M Storage Server to each system. This is referred to as an Elastic Configuration.

3.5.7 Licensing models and Elastic Expansion of Exadata on OD@Azure

Fully elastic expansion of Exadata on OD@Azure, allows customers to provision the database compute and storage resources for any workload and provides the scale for any size database DB Systems.

Exadata on OD@Azure provides Multi-VM-Clustering and Node-Subsetting capabilities to provide workload segregation and resource isolation on a single OD@Azure infrastructure. The features are as follows:

- Provision VM-cluster on a Subset of DB servers
- Expand an Existing VM Cluster by Adding VMs
- Shrink an Existing VM Cluster by Removing VMs

OD@Azure offers elastic infrastructure shapes to support workloads of different sizes. Customers can deploy flexible shapes that range from a Quarter Rack, 2 database and 3 storage servers, up to 32 database and 64 storage servers to meet a variety of CPU processing and storage requirements. Hence making it a perfect solution to consolidate on.

There are two licensing models:

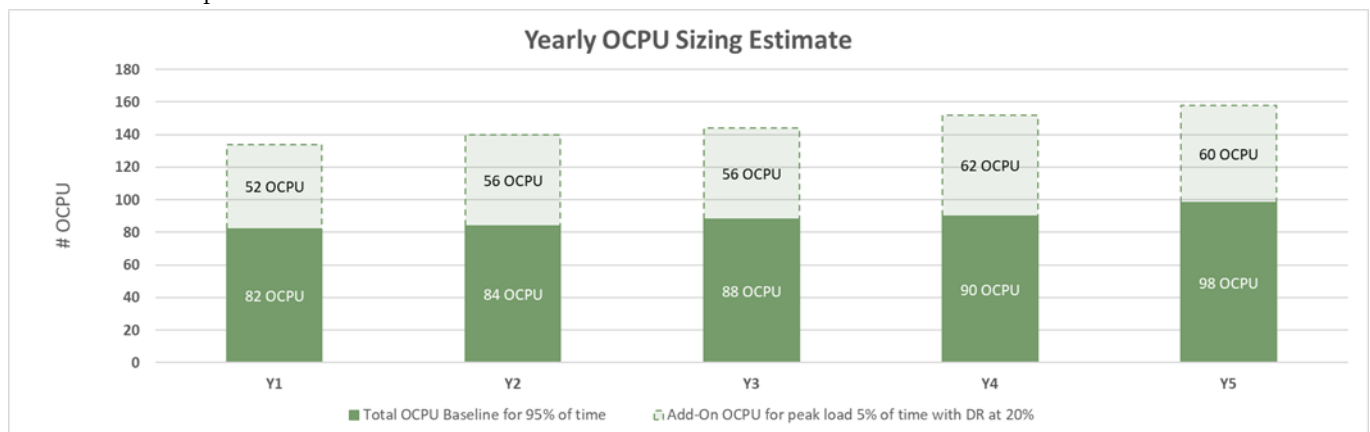
- Bring your own license (BYOL). Customer is only allowed to use the existing option licensed for.
- License included. This includes all RDBMS options.

The licensing model is set at Virtual Cluster level and can be changes at any point during the duration of the subscription. This is very effective in scenarios where more licenses are required due to the scope growth of the project and/or customer has a desire not to maintain on-premises licenses.

Example:

This is the proposed Consumption Plan over five years based on the sizing, growth and other assumptions.

Yearly OCPU Sizing Estimate	Y1	Y2	Y3	Y4	Y5
Total OCPU Baseline for 95% of time	82	84	88	90	98
Add-On OCPU for peak lead 5% of time with DR at 20%	52	56	56	62	60



sizing

3.5.8 Exadata Database@Azure Bill of Materials

Product	Metric	Qty
B93380 - Exadata Cloud Infrastructure - Quarter Rack - X9M	N/A	2
B88592 - OCI - Database Exadata OCPU	OCPU BYOL	152
B91627 - Object Storage - Requests#[0-5]	10,000 Requests per Month (first 50,000 free)	1
B91628 - Object Storage - Storage#[10-999999999]	Gigabyte Storage Capacity Per Month	1000

Annex

4.1 Security Guidelines

4.1.1 Oracle Security, Identity, and Compliance

Oracle Cloud Infrastructure (OCI) is designed to protect customer workloads with a security-first approach across compute, network, and storage – down to the hardware. It's complemented by essential security services to provide the required levels of security for your most business-critical workloads.

- [Security Strategy](#) – To create a successful security strategy and architecture for your deployments on OCI, it's helpful to understand Oracle's security principles and the OCI security services landscape.
- The [security pillar capabilities](#) reflect fundamental security principles for architecture, deployment, and maintenance. The best practices in the security pillar, help your organization to define a secure cloud architecture, identify and implement the right security controls, and monitor and prevent issues such as configuration drift.

4.1.2 References

- The Best Practices Framework for OCI provides architectural guidance about how to build OCI services in a secure fashion, based on recommendations in the [Best practices framework for Oracle Cloud Infrastructure](#).
- Learn more about [Oracle Cloud Security Practices](#).
- For detailed information about security responsibilities in Oracle Cloud Infrastructure, see the [Oracle Cloud Infrastructure Security Guide](#).

4.1.3 Compliance and Regulations

Cloud computing is fundamentally different from traditional on-premises computing. In the traditional model, organizations are typically in full control of their technology infrastructure located on-premises (e.g., physical control of the hardware, and full control over the technology stack in production). In the cloud, organizations leverage resources and practices that are under the control of the cloud service provider, while still retaining some control and responsibility over other components of their IT solution. As a result, managing security and privacy in the cloud is often a shared responsibility between the cloud customer and the cloud service provider. The distribution of responsibilities between the cloud service provider and the customer also varies based on the nature of the cloud service (IaaS, PaaS, SaaS).

4.1.4 Additional Resources

- [Oracle Cloud Compliance](#) – Oracle is committed to helping customers operate globally in a fast-changing business environment and address the challenges of an evermore complex regulatory environment. This site is a primary reference for customers on the Shared Management Model with Attestations and Advisories.
- [Oracle Security Practices](#) – Oracle's security practices are multidimensional, encompassing how the company develops and manages enterprise systems, and cloud and on-premises products and services.
- [Oracle Cloud Security Practices](#) documents.
- [Contract Documents](#) for Oracle Cloud Services.
- [OCI Shared Security Model](#)

- [OCI Cloud Adoption Framework Security Strategy](#)
- [OCI Security Guide](#)
- [OCI Cloud Adoption Framework Security chapter](#)

4.2 Networking Requirement Considerations

The below questions help to identify networking requirements.

4.2.1 Application Connectivity

- Does your application need to be exposed to the internet?
- Does your solution on DC (on-prem) need to be connected 24x7 to OCI in a Hybrid model?
 - Site-to-Site IPSEC (Y/N)
 - Dedicated Lines (FC) (Y/N)
- Are there any specific network security requirements for your application? (No internet, encryption, etc, etc)
- Will your application require connectivity to other cloud providers?
 - Site-to-Site IPSEC (Y/N)
 - Dedicated Lines (FC) (Y/N)
- Will your application require inter-region connectivity?
- Are you planning to reuse IP addresses from your on-premises environment in OCI?
- If yes, what steps have you taken to ensure IP address compatibility and avoid conflicts?
- How will you handle network address translation (NAT) for IP reuse in OCI?
- Will you bring your own public IPs to OCI?

4.2.2 DR and Business Continuity

- Does your organization need a Business Continuity/DR Plan to address potential disruptions?
 - Network Requirements (min latency, bandwidth, etc)
 - RPO/RTO values
- What are your requirements regarding Data Replication and Geo-Redundancy (different regions, restrictions, etc.)?
- Are you planning to distribute incoming traffic across multiple instances or regions to achieve business continuity?
- What strategies do you require to guarantee minimal downtime and data loss, and to swiftly recover from any unforeseen incidents?

4.2.3 High Availability and Scalability

- Does your application require load balancing for high availability and scalability? (y/n)
 - Does your application span around the globe or is regionally located?
 - How do you intend to ensure seamless user experiences and consistent connections in your application (session persistence, affinity, etc.)?
 - What are the network Security requirements for traffic management (SSL offloading, X509 certificates management, etc.)?
 - Does your application use name resolutions and traffic steering across multiple regions (Public DNS steering)?

4.2.4 Security and Access Control

- Some of the below questions help you to adopt the right sizing and deployment model of the network firewall.
 - Does the customer need to protect traffic from VCN to VCN?
 - Does the customer need to protect traffic from subnet to subnet in the same VCN?
 - When deploying an OCI Network Firewall in a dedicated HUB or secure VCN, do you want to protect inter-VCN traffic and/or inter-subnet traffic from within the same VCN?
 - Does the customer need to protect incoming or egressing traffic to the internet?

- Does the customer need to protect internal traffic (including on-premises via IPSEC/FC)?
 - Is the network performance critical?
 - Does the customer have any requirement on network isolation (i.e., internet traffic never traverses or is mixed with internal traffic)?
- Have you considered the importance of protecting your web applications from potential cyber threats using a Web Application Firewall (WAF)?

4.2.5 Monitoring and Troubleshooting

- How do you plan to monitor your application's network performance in OCI?
- How can you proactively address and resolve any potential network connectivity challenges your company might face?
- How do you plan to troubleshoot your network connectivity?

4.3 Networking Solutions

4.3.1 OCI Network Firewall

Oracle Cloud Infrastructure Network Firewall is a next-generation managed network firewall and intrusion detection and prevention service for your Oracle Cloud Infrastructure VCN, powered by Palo Alto Networks®.

- [Overview](#)
- [OCI Network Firewall](#)

4.3.2 OCI Load Balancer

The Load Balancer service provides automated traffic distribution from one entry point to multiple servers reachable from your virtual cloud network (VCN). The service offers a load balancer with your choice of a public or private IP address and provisioned bandwidth.

- [Load Balancing](#)
- [Overview](#)
- [Concept Overview](#)

4.3.3 OCI DNS Traffic Management

Traffic Management helps you guide traffic to endpoints based on various conditions, including endpoint health and the geographic origins of DNS requests.

- [Concept Overview](#)
- [DNS](#)

4.3.4 OCI WAF

Protect applications from malicious and unwanted internet traffic with a cloud-based, PCI-compliant, global web application firewall service.

- [Cloud Security Web Application Firewall](#)
- [Add WAF to a load balancer](#)

4.3.5 OCI IGW

An internet gateway is an optional virtual router that connects the edge of the VCN with the internet. To use the gateway, the hosts on both ends of the connection must have public IP addresses for routing

- [Managing IGW](#)

4.3.6 OCI Site-to-Site VPN

Site-to-site VPN provides a site-to-site IPSec connection between your on-premises network and your virtual cloud network (VCN). The IPSec protocol suite encrypts IP traffic before the packets are transferred from the source to the destination and decrypts the traffic when it arrives. Site-to-Site VPN was previously referred to as VPN Connect and IPSec VPN.

- [Overview IPSec](#)
- [Setup IPSec](#)

4.3.7 OCI Fast Connect

FastConnect allows customers to connect directly to their Oracle Cloud Infrastructure (OCI) virtual cloud network via dedicated, private, high-bandwidth connections.

- [FastConnect](#)
- [Concept Overview](#)

4.3.8 OCI VTAP

A Virtual Test Access Point (VTAP) provides a way to mirror traffic from a designated source to a selected target to facilitate troubleshooting, security analysis, and data monitoring

- [VTAP](#)
- [Network VTAP Wireshark](#)

4.3.9 OCI NPA

Network Path Analyzer (NPA) provides a unified and intuitive capability you can use to identify virtual network configuration issues that impact connectivity. NPA collects and analyzes the network configuration to determine how the paths between the source and the destination function or fail.

- [Path Analyzer](#)

4.3.10 OCI DRG (Connectivity Options)

A DRG acts as a virtual router, providing a path for traffic between your on-premises networks and VCNs, and can also be used to route traffic between VCNs. Using different types of attachments, custom network topologies can be constructed using components in different regions and tenancies.

- [Managing DRGs](#)
- [OCI Pilot Light DR](#)
- [Peering VCNs in different regions through a DRG](#)

4.3.11 OCI Oracle Cloud Infrastructure Certificates

Easily create, deploy, and manage Secure Sockets Layer/Transport Layer Security (SSL/TLS) certificates available in Oracle Cloud. In a flexible Certificate Authority (CA) hierarchy, Oracle Cloud Infrastructure Certificates help create private CAs to provide granular security controls for each CA.

- [SSL TLS Certificates](#)

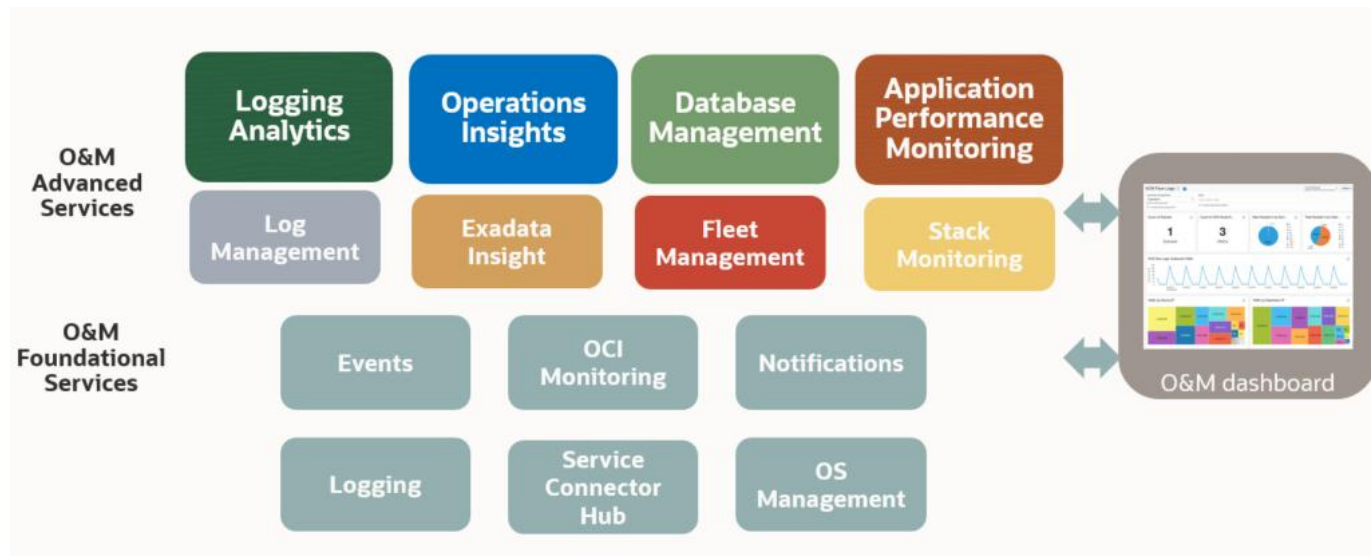
4.3.12 OCI Monitoring

You can monitor the health, capacity, and performance of your Oracle Cloud Infrastructure resources by using metrics, alarms, and notifications. For more information, see [Monitoring](#) and [Notifications](#).

- [Networking Metrics](#)

4.4 Manageability

OCI offers a full set of services to cover all Observability and Monitoring requirements.



OCI Observability

Thanks to AI algorithms the OCI O&M (Observability and Management) solutions offer valuable insights into system status, requirements, and trends. Furthermore, it identifies SQL performance issues. This proactive approach empowers proactive measures to prevent future issues.

4.4.1 OCI O&M Services List

The observability and management services include the following services:

[Application Performance Monitoring](#) offers in-depth insight into application performance and facilitates rapid diagnostics to ensure a reliable level of service. This includes monitoring various components and application logic spread across clients, third-party services, and backend computing tiers, whether on-premises or in the cloud.

[Database Management](#) provides comprehensive database performance diagnostics and management capabilities to monitor and manage Oracle databases.

[Logging](#) lets you enable, view, and manage all the logs in your tenancy and provides access to logs from Oracle Cloud Infrastructure resources. These logs include critical diagnostic information that describes how resources are performing and being accessed.

[Logging Analytics](#) is a unified, integrated cloud solution that enables users to monitor, aggregate, index, analyze, search, explore, and correlate all log data from their applications and system infrastructure.

[OCI Monitoring](#) enables you to query [metrics](#) and manage [alarms](#). Metrics and alarms help monitor the health, capacity, and performance of your cloud resources.

[Ops Insights](#) provides a 360-degree insight into the resource utilization and capacity of Oracle Autonomous Databases. You can easily analyze CPU and storage resources, forecast capacity issues, and proactively identify SQL performance issues across a fleet of Autonomous Databases.

[Service Connector Hub](#) is a cloud message bus platform that offers a single pane of glass for describing, running, and monitoring interactions for data moving between Oracle Cloud Infrastructure services.

[Stack Monitoring](#) enables proactive monitoring of applications and their underlying stack, including application servers and databases. By discovering all components of an application, including the application topology, Stack Monitoring automatically collects status, load, response, error, and utilization metrics for all application components. Each component of the application stack is referred to as a resource.

4.4.2 Real-Time Monitoring Annex

Service/Product Name	Description	Collateral
Monitoring	OCI Monitoring collects PaaS and IaaS OCI services metrics. It is enabled by default for all the OCI services.	List of metrics collected by default
OCI Application Performance Monitor	APM is a Distributed Tracing System as a Service. It enables DevOps teams to follow every step of every task. It uses open standards such as OpenTelemetry to monitor various programming languages. Plus, it includes a dedicated Java agent to track older J2EE applications, ensuring complete transaction tracing even in mixed environments.	OCI Application Performance Monitoring
OCI Console	The Service Console offers a list of visual representations and basic information about critical metrics like CPU, memory, and storage.	OCI Console Resource Usage Tracking
OCI Database Management (opt to OEM)	It is an OCI-managed service that simplifies database operations and enhances efficiency. It offers advanced monitoring and diagnostic capabilities, enabling proactive management and optimization of database performance.	List of metrics collected by OCI Database Management
Stack Monitoring	Stack Monitoring lets you proactively monitor an application and its underlying application stack, including application servers and databases.	Stack Monitoring for Oracle Database
Third-Party Tools - Service Connector Hub	OCI provides complete O&M capabilities. However, for customers who prefer to use their own tools, OCI allows seamless integration through the Service Connect Hub.	OCI Connector Hub Third-Party Tools Use Cases

4.4.3 Performance and Tuning Annex

Service/Product Name	Description	Collateral
OCI Logging	The OCI Logging service is a highly scalable and fully managed single pane of glass for all the logs in your tenancy. Logging provides access to logs from Oracle Cloud Infrastructure resources. These logs include critical diagnostic information that describes how resources are performing and being accessed.	OCI Logging
OCI Monitoring	Use the Oracle Cloud Infrastructure Monitoring service to actively and passively monitor cloud resources using the Metrics and Alarms features. Metric data posted to the Monitoring service is only presented to you or consumed by the Oracle Cloud Infrastructure features that you enable to use metric data.	OCI Monitoring
OCI Dashboard	The Console Dashboards service allows you to create custom dashboards in the Oracle Cloud Infrastructure Console to monitor resources, diagnostics, and key metrics for your tenancy.	OCI Dashboard
OCI Logging Analytics	OCI Logging Analytics empowers users to analyze log data from diverse sources across their infrastructure. It provides insights into system performance, identifies trends, and enables proactive resource optimization by correlating data from multiple layers of the infrastructure.	OCI Logging Analytics
OCI Application Performance Monitor	APM allows to drill down from user sessions till the single DB query or external call to identify performance bottleneck.	OCI Application Performance Monitoring
OCI Database	Is an OCI-managed service that offers performance and tuning capabilities. It	Database Management

Service/Product Name	Description	Collateral
Management - PerfHub	provides the same performance and tuning features as the Oracle Enterprise Manager (OEM) Performance and Tuning Pack but in a managed solution.	Performance Hub
Ops Insights Sql Warehouse and Capacity Planning	OCI Ops Insights allows for the tracking of metrics charts and data collection. It allows for the correlation of resources across various infrastructure layers. Additionally, it predicts high resource utilization for computing and database instances.	OCI Operations Insight SQL Warehouse OCI Operations Insight Capacity planning

4.4.4 Administration Annex

Service/Product Name	Description	Collateral
OCI Console	The OCI Console is embedded in all cloud services. It allows basic tasks such as listing, starting, stopping, or termination of resources.	OCI Console
OCI Database Management	This OCI-managed service allows you to manage your databases. It provides a subset of functionalities offered by the OEM.	Database Management
OCI Organization Management	The OCI Console has several tenancy management features. You can use Organization Management to centrally manage your multi-tenancy environment.	Organization Management

4.4.5 Troubleshooting Annex

Service/Product Name	Description	Collateral
Logging Analytics	OCI Logging Analytics can handle log events generated by all software applications and infrastructure on the cloud or on-premises. For Oracle software logs, a predefined severity pre-classification exists based on Oracle experience.	OCI Logging Analytics OCI Logging Analytics for Exa
OCI Application Performance Monitor	APM allows to drill down from user sessions till the application logs to find the root cause.	OCI Application Performance Monitoring
OCI Database Management	OCI-managed service that allows you to drill down and correlate metrics and data from different layers. it provides built-in links that allow you to connect to other O&M services (ex. Ops Insights).	Database Management
Ops Insights	OCI Ops Insights allows tracking of metrics charts and data collection. It allows for the correlation of resources from different infrastructure layers.	OCI Operations Insight OCI ExaInsight

4.4.6 Cost Control and Chargeback Annex

Service/Product Name	Description	Collateral
Ops Insights Capacity Planning	This OCI-managed service allows one to predict the resource consumption for a year. With tags, you can associate the forecast and the consumption to a specific department.	Operations Insight Capacity Planning
Cost Analysis	Cost Analysis is an easy-to-use visualization tool to help you track and optimize your Oracle Cloud Infrastructure spending. It allows for the generation of charts and the download of accurate and reliable tabular reports of aggregated cost data. With tags, you can associate the forecast and the consumption to a specific department.	OCI Cost Analysis
Usage RestAPI	OCI offers various RestAPI's to manage services, including the one for cost management.	OCI Usage RestAPI

4.5 Onboarding

Oracle offers a comprehensive onboarding program for new customers, designed to facilitate their initial journey within the OCI (Oracle Cloud Infrastructure) environment. The [OCI Digital Onboarding Home](#) is an excellent starting point for customers to

familiarize themselves with OCI. This resource provides a self-guided learning path covering essential aspects such as initial access, cloud console navigation, fundamental resource creation, and implementing security best practices.

The onboarding process continues with a self-service kick-off workshop, tailored for EMEA customers, which includes an extensive playbook. This workshop delves deeper into various topics:

- An introduction to OCI and its features.
- A step-by-step guide to getting started.
- A guided tour of the platform.
- An overview of live lab sandbox environments for hands-on practice.
- Comprehensive best practices encompassing security, reliability, performance optimization, cost management, and monitoring strategies.
- A support guide to assist customers with any queries or issues.
- A satisfaction survey to gather feedback and ensure an optimal user experience.

This structured approach ensures that new customers receive a well-rounded introduction to OCI, empowering them to make the most of their new tenancy.